ARC RESEARCH · AI UPDATE

Neoclouds & Neolabs

Why AI compute is being rebuilt outside Big Tech, how it works, and what could break

Data as of 6 October 2026 · Educational purposes only, not investment advice

AGENDA

What's in this note

1 **Why we're here** — Demand, spending, GPU prices — **Slides 4–7**

2 **Neoclouds** — How GPU clouds work and are financed — **Slides 8–23**

3 **Neolabs** — How new labs raise and spend — **Slides 24–44**

4 **Risks & challenges** — What could break, technically — **Slides 45–61**

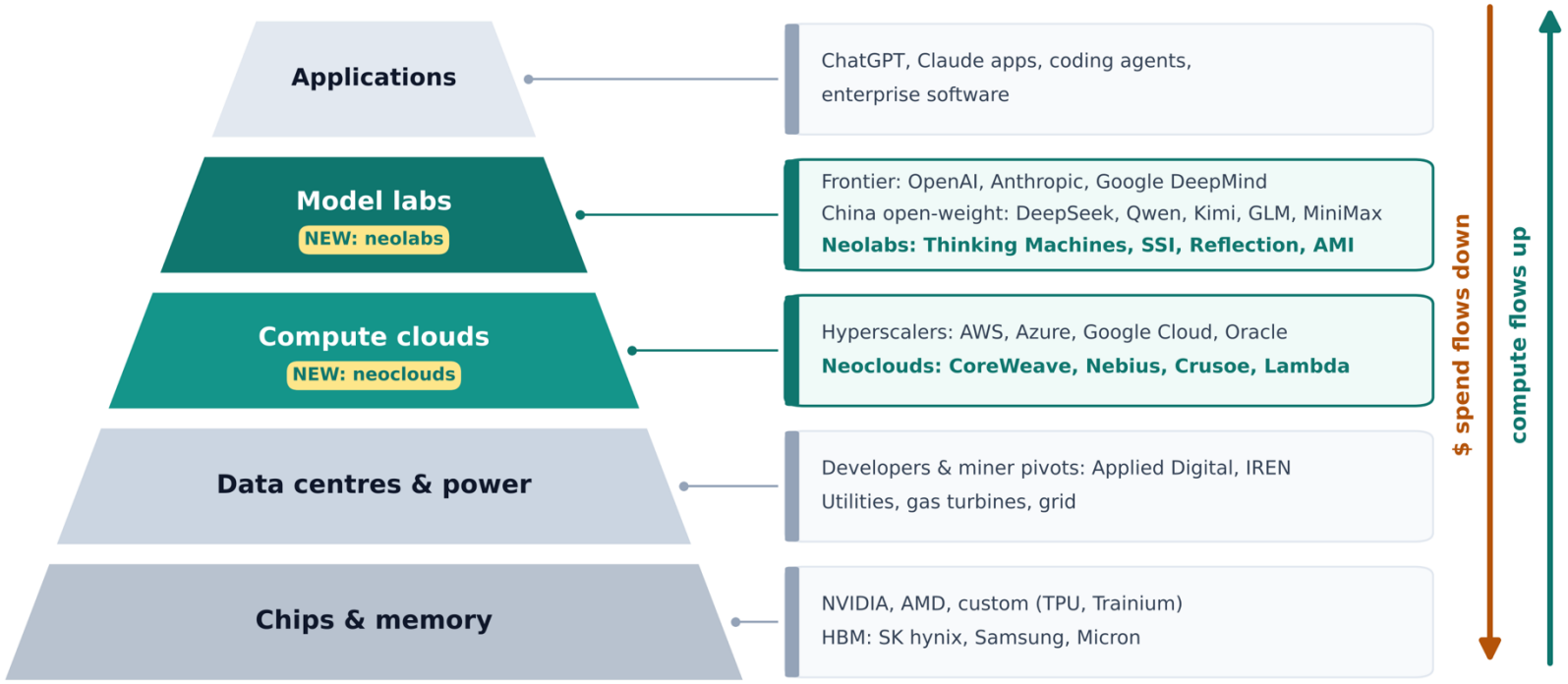
Start: where they sit (slide 3) · End: ARC view and sources (slides 62–63)

Two new layers in the AI stack

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Exhibit 1: The AI compute stack: where neoclouds and neolabs fit

Neoclouds rent GPU capacity; neolabs rent it to train models. Money flows down the stack, compute flows up



Schematic, not to scale. Company examples are illustrative, not exhaustive. China open-weight labs release model weights publicly (latest as of Oct 2026: Kimi K3, DeepSeek V4.1, GLM-5.3, Qwen 3.8, MiniMax M3). Neocloud and neolab layers highlighted. Sources for company roles: company filings and releases cited on later slides.

SO WHAT

Neoclouds: clouds built only to rent GPUs

Neolabs: new labs renting GPUs to train models

Both sit between NVIDIA and the apps

SECTION 1 OF 4

Why we're here

Demand, spending and the price of compute

COMING UP

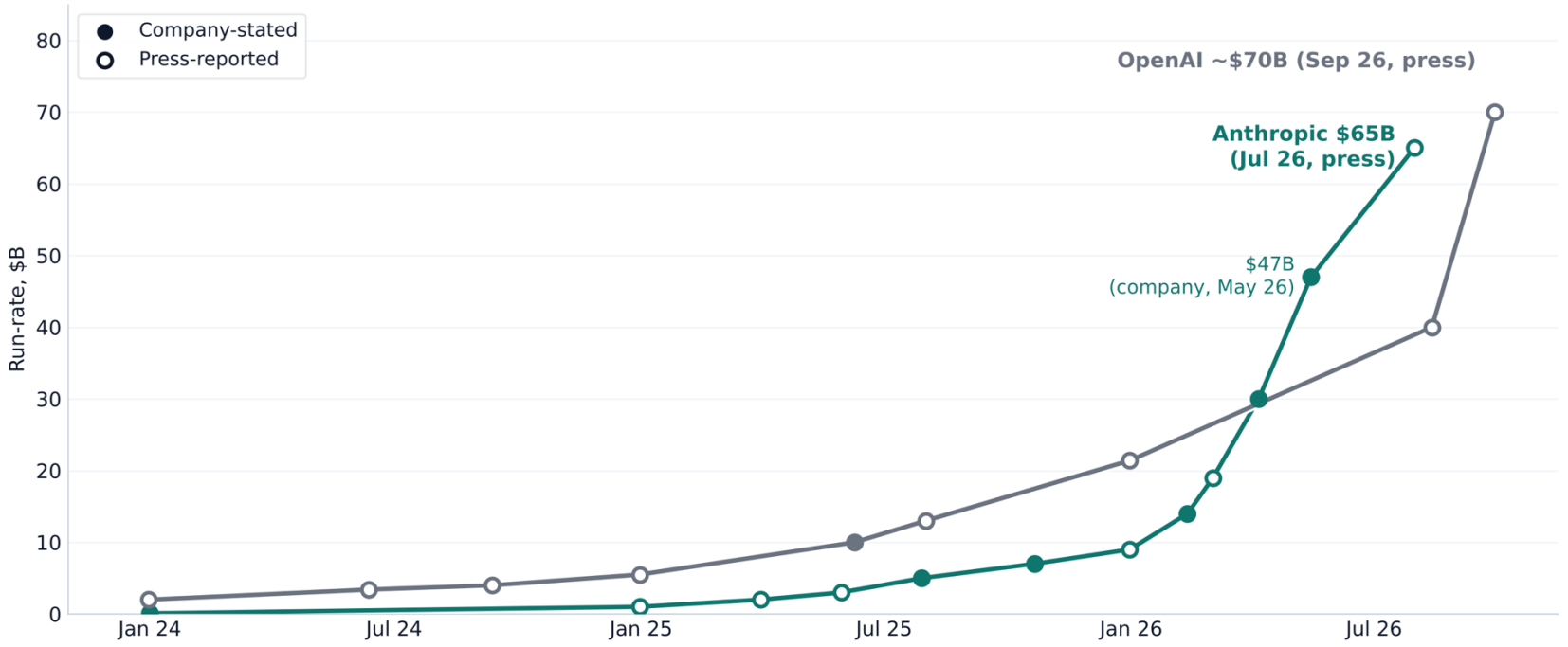
Lab revenue is compounding 5
Big Tech capex up 4.9x 6
GPU rental prices 7

Lab revenue is compounding at record speed

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Exhibit 2: Annualised revenue run-rate, Anthropic and OpenAI ($B)

Both labs crossed ~$65-70B run-rate in 2026, up from ~$1B and ~$2B two years earlier



Source: Epoch AI revenue-reports dataset (row-level links); Anthropic news posts (Series F, G, H; Oct 2025 statement; Apr 2026 compute post); press: CNBC, The Information, Bloomberg, FT, NYT, WSJ, Axios. Solid marker = company-stated; hollow = press-reported. Run-rate = annualised current revenue, not GAAP. Both firms are private; no audited figures.

SO WHAT

Anthropic: ~$1B to $65B run-rate in 19 months

OpenAI: ~$70B run-rate by Sep 2026

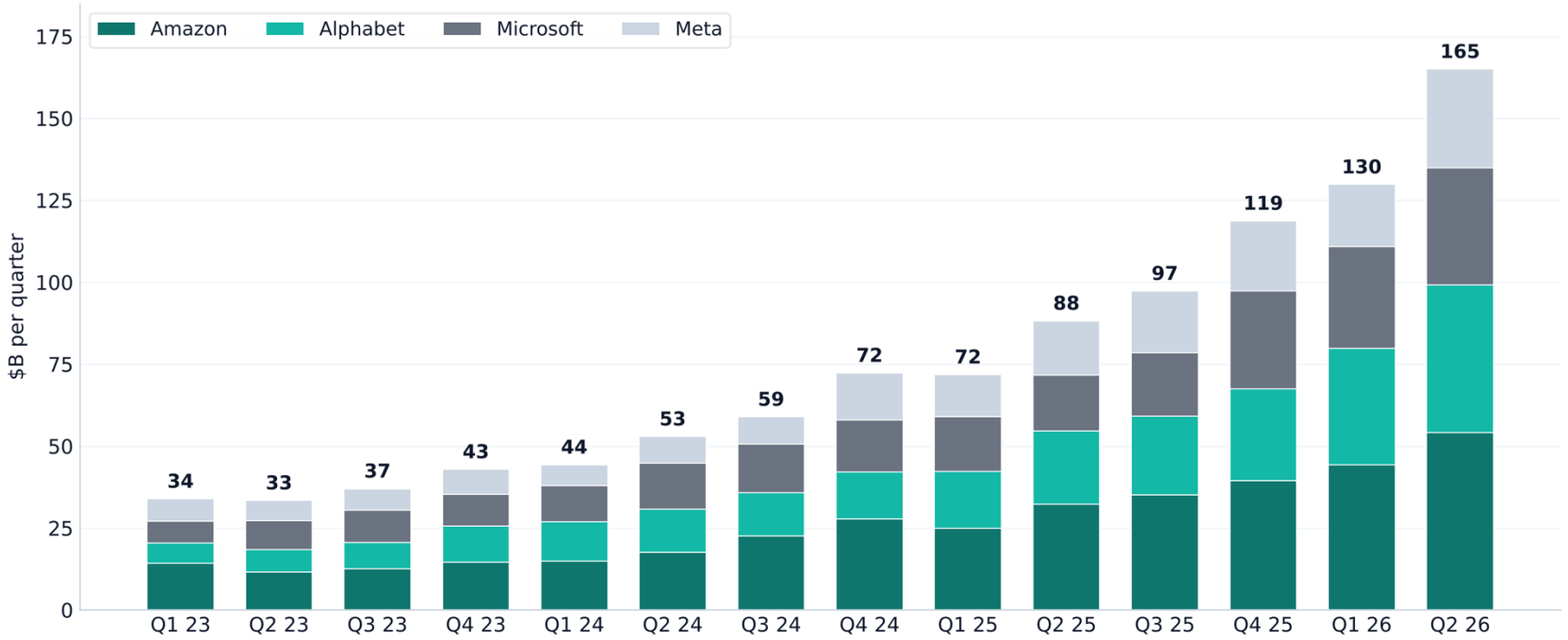
Paying customers fund the compute bill

Big Tech now spends $165B a quarter

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Exhibit 3: Quarterly capex, Microsoft, Alphabet, Amazon, Meta ($B, calendar qtr)

Combined spend rose from $34B a quarter in Q1 2023 to $165B in Q2 2026 (4.9x)



Source: company earnings releases / cash-flow statements via stockanalysis.com; Q2 2026 (and Amazon, Alphabet Q1 2026) checked against primary releases. Cash basis: purchases of property & equipment (Microsoft: additions to PP&E); excludes finance leases, so below Microsoft/Meta headline 'capex incl. leases'. Amazon gross of sale proceeds. Microsoft FY mapped to calendar. Totals = sum, ours.

SO WHAT

4.9x since Q1 2023

Meta guides $130-145B for 2026

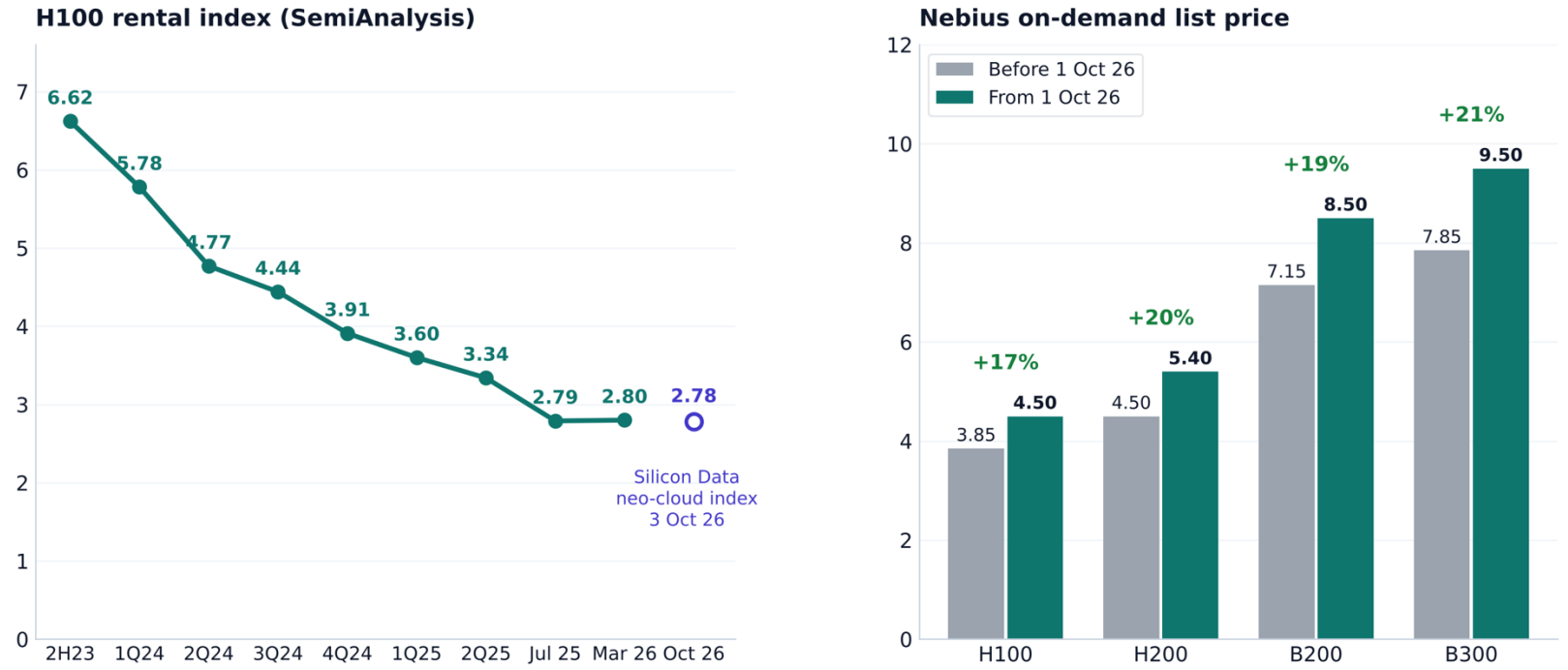
Even so, Meta and Microsoft rent from neoclouds

Old GPUs got cheaper; new ones are scarce

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Exhibit 4: H100 rental index vs Nebius list prices ($ per GPU-hour)

Old-gen H100 rents fell ~58% then flattened near $2.80; newest-gen list prices were raised 17-21% on 1 Oct 2026



Source: left, SemiAnalysis H100 GPU rental index (spot-contract composite, gpu-index.semianalysis.com); Silicon Data H100 Rental Index, neo-cloud segment (3 Oct 2026). The two indices measure different things and are not spliced. Right: nebius.com/prices, on-demand list before and effective 1 Oct 2026. List prices sit above market index rates. % change = after/before - 1, ours.

SO WHAT

H100 rent –58% since 2H23, flat since mid-25

Nebius list prices +17-21% on 1 Oct

CoreWeave: new deals at ~$40M per MW

SECTION 2 OF 4

Neoclouds

Clouds built only to rent GPUs

COMING UP

Neocloud vs hyperscaler 9–10

Capacity: live vs contracted 11

Inside CoreWeave, Nebius, Crusoe 12–16

Revenue and unit economics 17–18

Financing: who lends 19–21

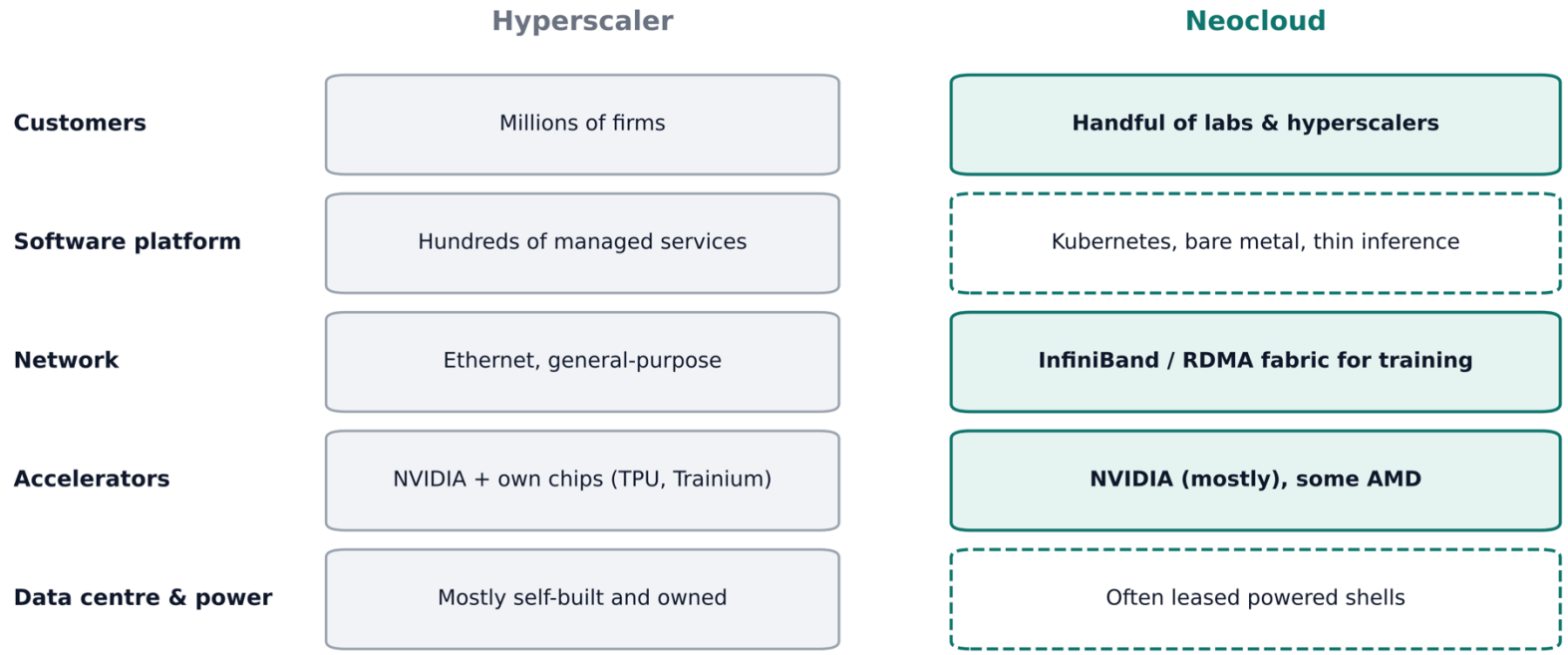
Valuation and who pays 22–23

A neocloud is a GPU-only cloud

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Exhibit 5: What a hyperscaler owns vs what a neocloud owns

A neocloud is a narrow, GPU-only cloud: it rents leased-power capacity to a few big buyers on multi-year contracts



Schematic, not to scale. Solid = layer the company builds/owns; dashed = leased or largely absent. Neocloud examples: CoreWeave leases sites from developers such as Applied Digital (400 MW at Polaris Forge 1, DCD); CoreWeave FY2025 largest customer ~67% of revenue (Microsoft); managed inference >$100M ARR (Q2 2026 call). Hyperscaler examples: AWS, Azure, Google Cloud. Generalisations; individual firms differ.

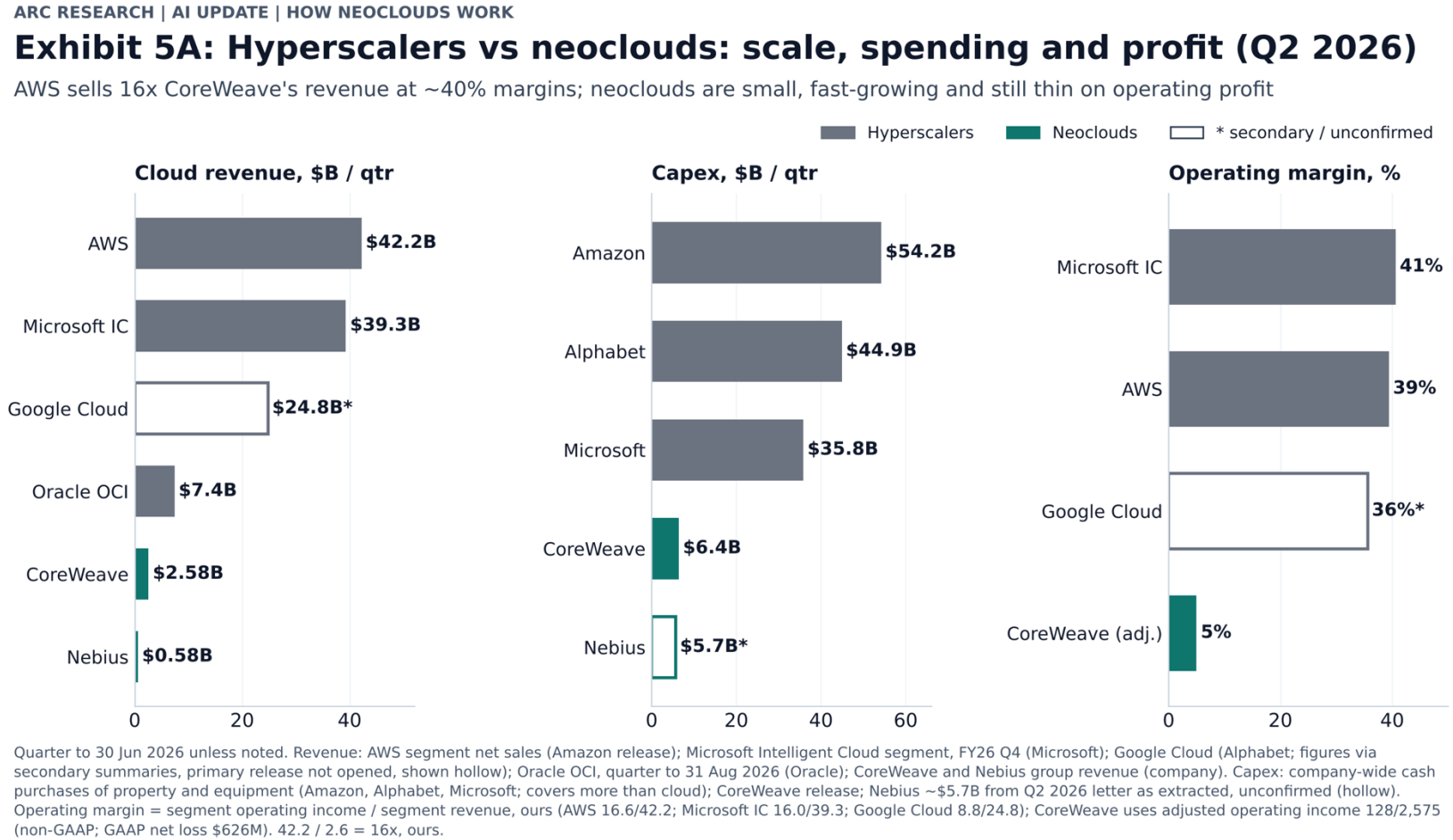
SO WHAT

Sells speed: power, GPUs, live cluster

Few buyers, multi-year take-or-pay

Thin software, deep networking

Neoclouds are a fraction of hyperscaler scale



SO WHAT

AWS revenue 16x CoreWeave's

Hyperscalers earn ~40% margins

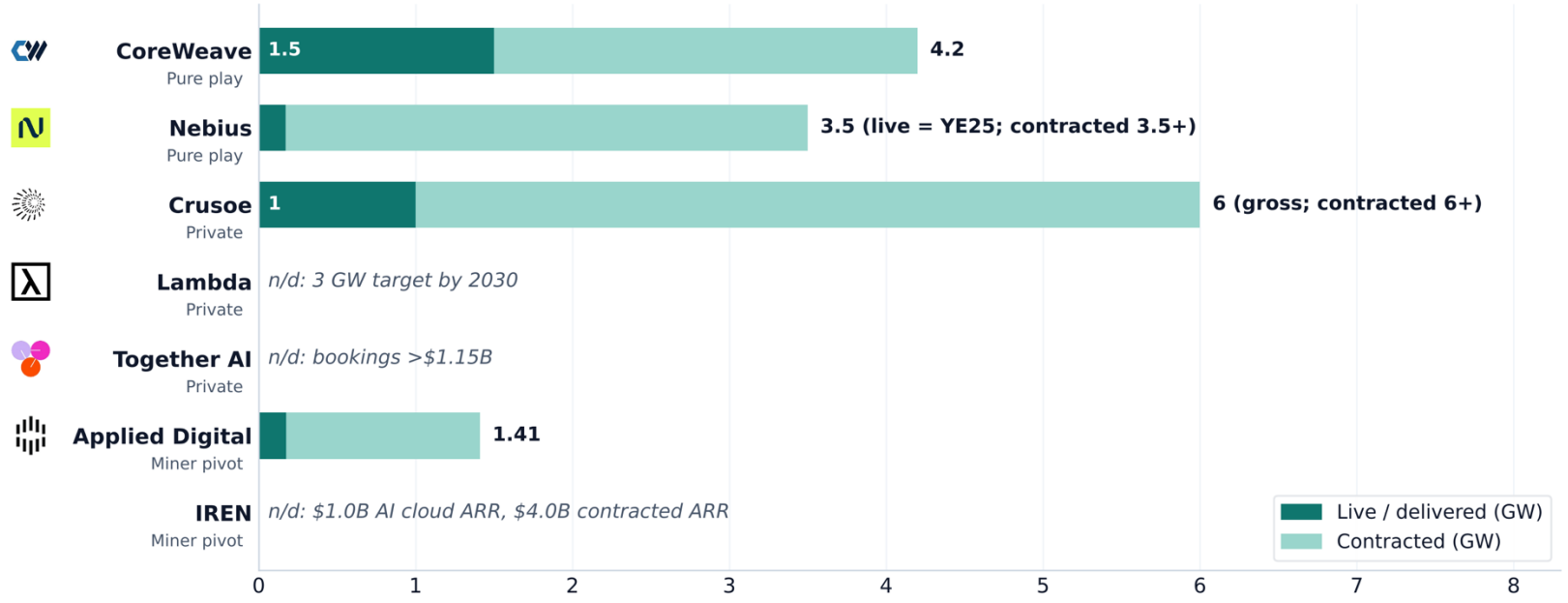
Neoclouds: fast growth, thin profit

Three kinds of neocloud, all short of power

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Exhibit 6: Neocloud capacity: live vs contracted power (GW)

Contracted power runs 3-6x what is live today: the build-out, not the demand, is the bottleneck



Source: CoreWeave Q2 2026 release & call (live ~1.5 GW; contracted 4.2 GW as of 11 Aug 2026); Nebius Q4 2025 letter (~170 MW connected YE2025) and Q1 2026 letter (>3.5 GW contracted); Crusoe Series F release, 17 Sep 2026 (1 GW gross delivered, 6+ GW gross contracted); Applied Digital FQ4 2026 release (175 MW live, 1,410 MW contracted critical IT). Dates differ by company. n/d = not disclosed: Lambda, Together AI and IREN do not report comparable GW figures; alternative company metrics shown instead.

SO WHAT

CoreWeave: 1.5 GW live, 4.2 GW contracted

Crusoe: 6+ GW contracted, $30.9B value

Bitcoin miners resell their power sites

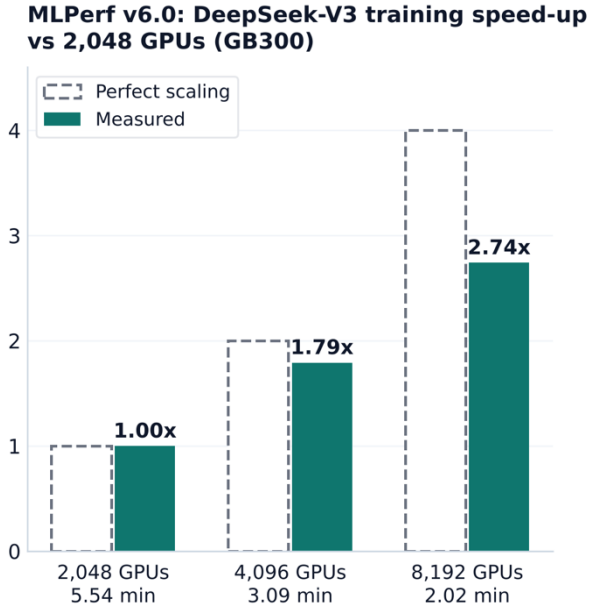
CoreWeave sells reliability at scale

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Exhibit 6A: CoreWeave: a bare-metal GPU stack plus fleet automation

Its edge is operational: no hypervisor, a training-grade network, and software that finds and swaps failing nodes automatically

Developer tools	Weights & Biases, OpenPipe (RL), Marimo
Mission Control	burn-in, health checks, auto-replace nodes
SUNK	Slurm job scheduler running on Kubernetes
CKS	Kubernetes on bare metal, no hypervisor
Network	InfiniBand fat-tree, 400 Gb/s per GPU, DPUs
Hardware	NVIDIA GB200 / GB300 NVL72 racks



Sources (company): docs.coreweave.com/products/cks (bare-metal Kubernetes, no hypervisor; BlueField DPU per node; non-blocking InfiniBand fat-tree with SHARP); GB200 NVL72 GA release (4 Feb 2025: 400 Gb/s per GPU, clusters up to 110,000 GPUs); Mission Control page and node-lifecycle blog (24-48 h burn-in, NCCL/burn tests, auto drain/RMA, 'up to 96% goodput' claim); MLPerf Training v6.0 release (16 Jun 2026): DeepSeek-V3 671B in 5.54 / 3.09 / 2.02 min on 2,048 / 4,096 / 8,192 GB300 GPUs. W&B price ~$1.7B reported by The Information (press). Speed-up = 5.54 / time; ideal = GPU multiple, ours.

KEY NUMBERS

Fleet
GB200/GB300 NVL72; Rubin validated

Capacity
~1.5 GW live / 4.2 GW contracted

List price
H100 $6.16, GB200 $10.50 /GPU-hr

Contracts
$104B RPO; OpenAI $22.4B, Meta $35B*

Financing
~$35B debt; $3.7B convert (Sep)

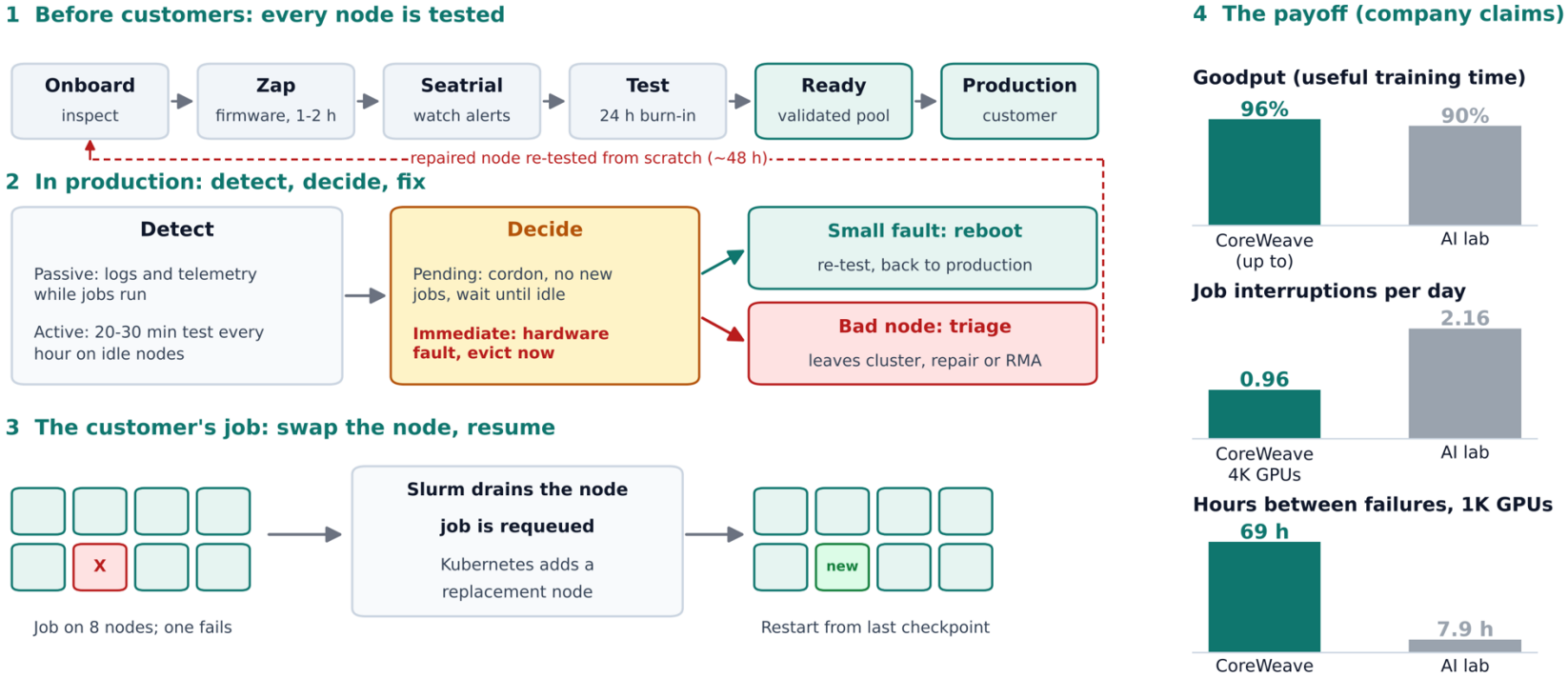
Sites
51 data centres

How CoreWeave swaps out failing nodes

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Exhibit 6B: How CoreWeave finds and swaps a failing node

Automated checks catch the bad node, pull it out and swap in a healthy one; the job restarts from its last checkpoint



Sources: CoreWeave docs (node lifecycle Day 0-2, state transitions, HPC Verification, CKS node pools and Prefill, SUNK fault tolerance and checkpoint/restart, NVL72 instances); CoreWeave blog 'Achieve AI infrastructure goodput of up to 96%' (27 Mar 2025): goodput and interruptions/day from CoreWeave tests and customers vs 'a leading AI lab'; MTTF for a 1K-GPU job, AI-lab study 7.9 h vs CoreWeave internal testing average 69 h (range 34-104 h). All three charts are company claims, not independently verified. SemiAnalysis ClusterMAX 3.0 (Sep 2026): Platinum rating, third in a row.

SO WHAT

Every node burned in for 24 h first

Idle nodes self-test every hour

Bad node out, new node in, job resumes

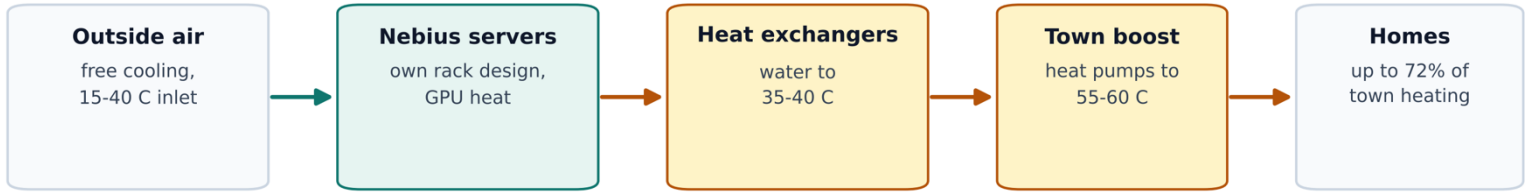
Nebius builds the whole stack itself

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Exhibit 6B: Nebius: full-stack, from its own servers to the town's heating

Designs its own servers, racks and buildings; runs free-cooled Finnish halls at PUE ~1.1 and sells the waste heat

Heat path at Mantsala, Finland



PUE ~1.1-1.13: for every 1 kW to the GPUs, only ~0.1 kW more for cooling and losses

(PUE 1.0 = no overhead; older enterprise halls run well above this)

Software on top

Soperator	Token Factory	Inferize (Oct 2026)
open-source Slurm on Kubernetes, auto GPU health checks	managed inference, 40+ models (launched Nov 2025)	cuts inference 'cold starts' (model load delays)

Sources (company): Nebius Finland release (8 Oct 2024): servers, racks and facility designed in-house, Mantsala 25 MW tripling to 75 MW, up to 60,000 GPUs, PUE 'as low as 1.1'; sustainability page: PUE 1.13 at full load, heat lifted to 35-40 C then boosted by the municipality to 55-60 C, up to 72% of the town's annual heating, 30+ GWh in two years; Soperator blog (open-source Slurm operator, Apache 2.0); Token Factory launch (5 Nov 2025); Inferize 6-K (1 Oct 2026). DCD (press): free-cooling inlet 15-40 C, company claim of 30-50% less server energy. PUE = total facility power / IT power. Schematic, not to scale.

KEY NUMBERS

Fleet
H100 to GB300; Rubin arriving

Capacity
0.8-1 GW live target; >5 GW contracted target

List price
H100 $4.50, B200 $8.50 /GPU-hr

Contracts
Microsoft $17.4-19.4B; Meta to $27B*

Financing
$8.5B debt; $5.75B converts (Aug)*

Sites
Finland, NJ, Kansas City, UK, Israel +

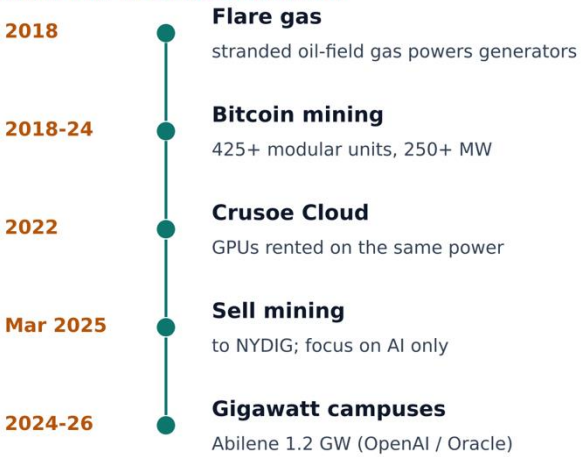
Crusoe starts with the power, not the GPUs

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Exhibit 6C: Crusoe: power first, then GPUs

Started by turning wasted oil-field gas into bitcoin mining; now builds gigawatt AI campuses with its own power behind the meter

How the business evolved



Abilene, Texas campus (design)



Sources: Crusoe newsroom: flare-gas mining (2021), NYDIG sale of mining + flare business (25 Mar 2025: 425+ modular units, 250+ MW), Abilene expansion (18 Mar 2025: 8 buildings, ~4M sq ft, 1.2 GW, up to 50,000 GB200 GPUs per building, direct-to-chip closed-loop liquid cooling, zero water evaporation, on-site solar + batteries, gas backup); GE Vernova: 29 LM2500XPRESS turbines ~35 MW each (~1 GW). Press: Epoch AI (Jul 2026) ~421 MW IT live at Abilene; DCD: Cheyenne 1.8 GW paused 10 Jun 2026 at the customer's request; Series F (17 Sep 2026). Campus drawing schematic, not to scale.

KEY NUMBERS

Fleet
GB200, B200, H100; AMD MI355X

Capacity
1 GW live / 6 GW contracted

List price
H100 $3.90, H200 $4.29 /GPU-hr

Contracts
$140B+ contracted; Abilene for OpenAI

Financing
$3.9B Series F; $11.6B Abilene JV*

Sites
Abilene 1.2 GW; TX, WY, Iceland, Norway

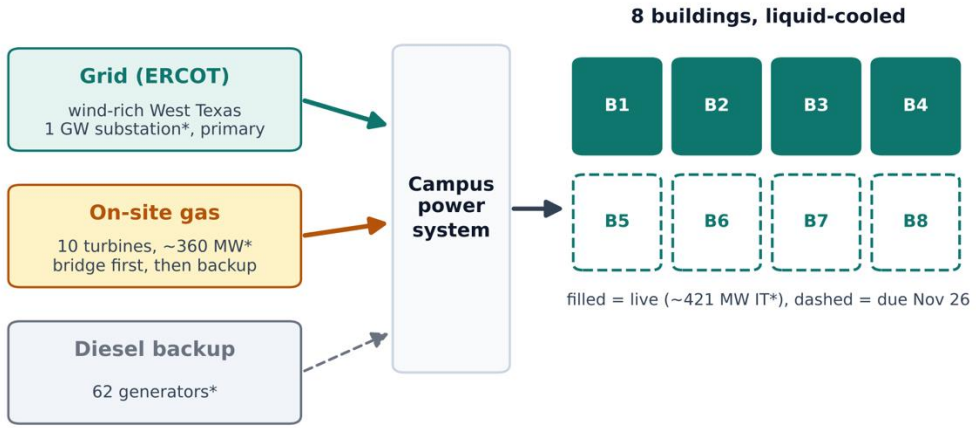
Crusoe brings its own power to skip the queue

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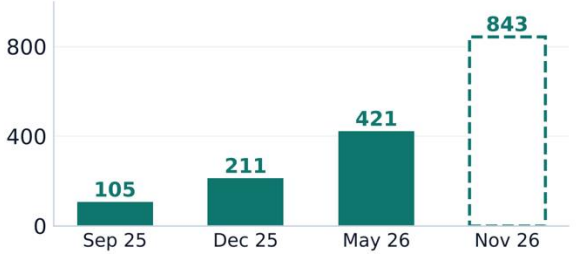
Exhibit 6E: How Crusoe turns power into GPUs

Go where energy is, bring your own gas to skip the grid queue, then let the grid take over and keep gas as backup

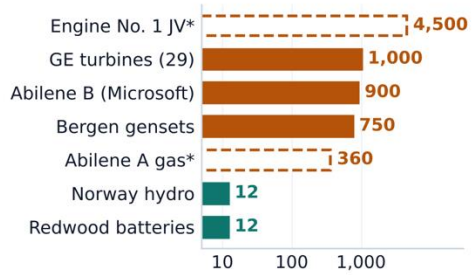
1 Abilene (1.2 GW, OpenAI/Oracle): how power reaches the GPUs



2 Abilene IT power, MW (Epoch est.*)



3 Own power lined up, MW (log scale)



The playbook

1 Go to the power

cheap, surplus energy
not Northern Virginia

2 Bring your own gas

turbines in weeks,
not a 4+ year queue

3 Grid takes over

gas stays as backup;
batteries smooth load

Sources: Crusoe blog 'BYO Power era' (17 Mar 2026: 'move the compute to the energy'; Abilene energized one year after groundbreaking); Crusoe Abilene blog (Aug 2025, curtailed wind); grid queue 4+ years: LBNL Queued Up; DCD (Jan 2025, ~360 MW gas plant, 10 turbines, TCEQ permit); Texas Tribune (Jul 2026, 62 diesel generators); Epoch AI satellite estimates of Abilene IT power (updated 5 Oct 2026; third party; Crusoe said 2 of 8 buildings operational on 9 Jun 2026); GE Vernova (Jul 2025, 29 LM2500XPRESS, nearly 1 GW); Crusoe releases: Bergen ~750 MW (2026), Abilene-Microsoft 900 MW on-site plant (27 Mar 2026), Redwood 12 MW / 63 MWh (Jun 2025), Polar Norway 12 MW (Jun 2025); Engine No. 1 JV 4.5 GW (DCD citing The Information, Mar 2025*). * = press-reported or third-party estimate.

SO WHAT

Own gas powers GPUs before the grid

Abilene live in ~1 year

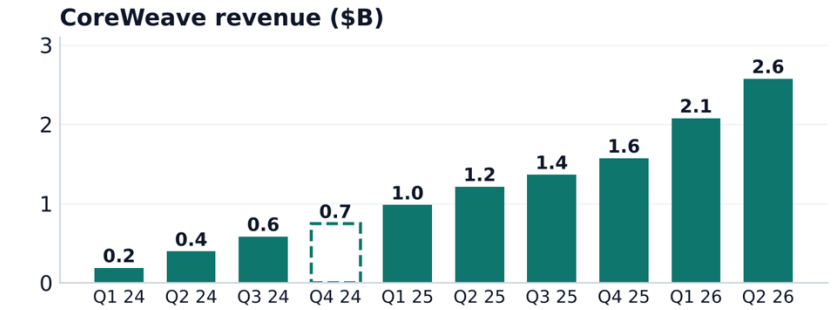
Trade-off: power-plant risk

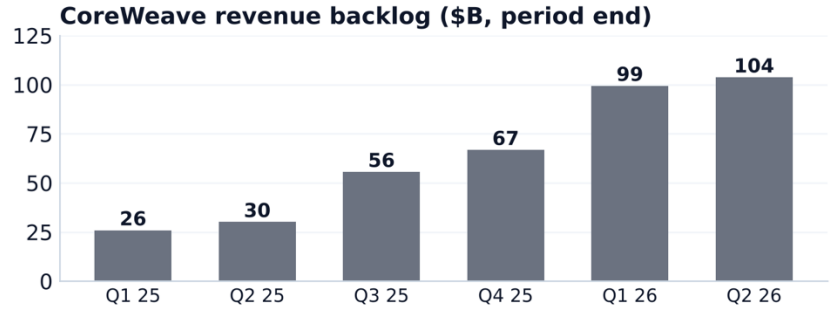
Revenue and backlog are surging

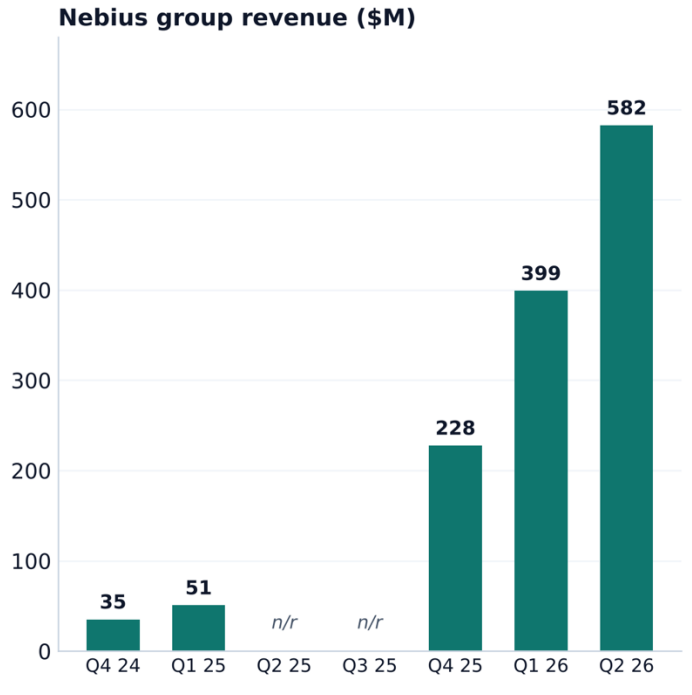
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Exhibit 7: CoreWeave and Nebius: quarterly revenue and CoreWeave backlog

CoreWeave revenue 13x in two years with $104B contracted; Nebius revenue 11x in five quarters







Source: CoreWeave quarterly earnings releases (Q1 2024-Q2 2026) and 10-Q RPO; Nebius shareholder letters (Q4 2025, Q1 2026, Q2 2026). Hollow bar = derived: CoreWeave Q4 2024 = FY2024 $1,915M less Q1-Q3 2024, ours. Backlog = company 'revenue backlog' (Q2 2026 = 10-Q RPO $103.7B). Nebius Q1-Q3 2024 and Q2-Q3 2025 not retrieved: shown as gaps, not estimated. Multiples: Q2 26 / Q2 24 and Q2 26 / Q1 25, ours.

SO WHAT

CoreWeave Q2: $2.6B, +112% y/y

$104B backlog: ~8x 2026 revenue

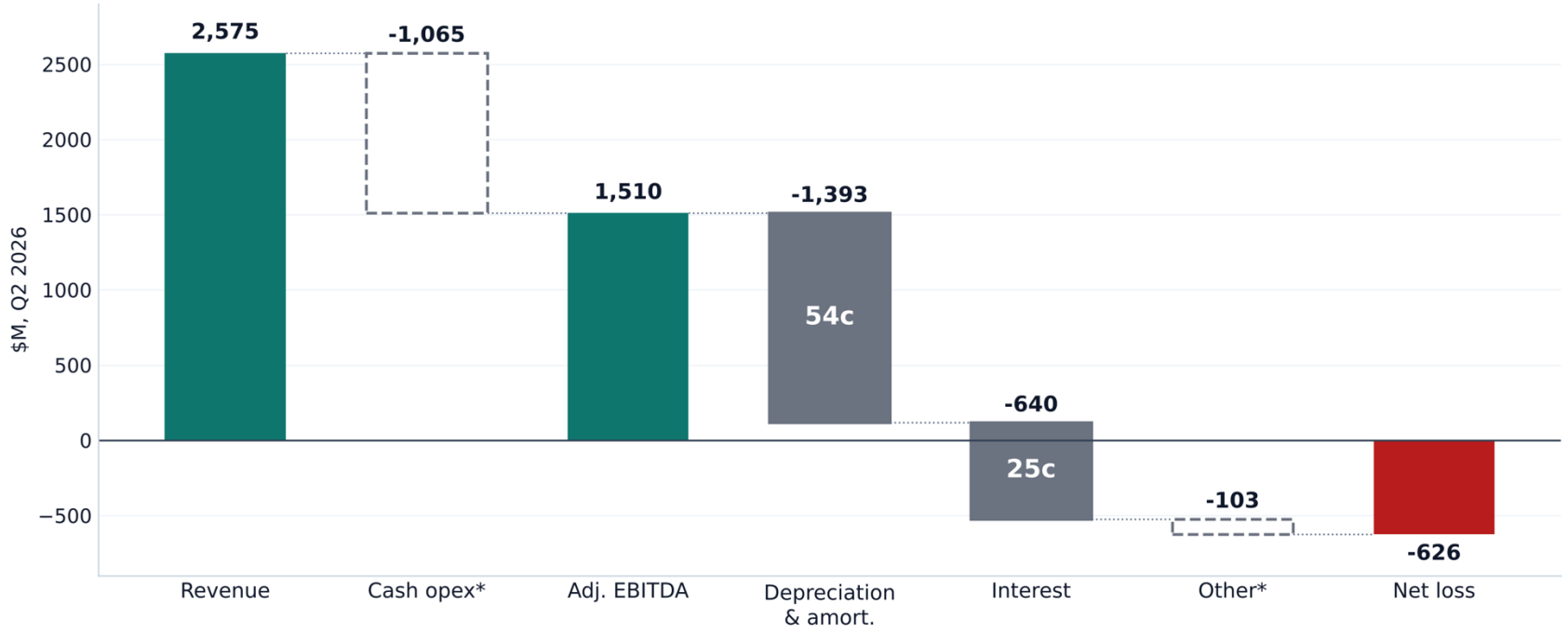
Nebius Q2: $582M, +454% y/y

High margins, eaten by depreciation and interest

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Exhibit 8: CoreWeave Q2 2026: from revenue to net loss ($M)

59% EBITDA margin, but depreciation takes 54c and interest 25c of every revenue dollar



Source: CoreWeave Q2 2026 earnings release (11 Aug 2026): revenue $2,575M, adjusted EBITDA $1,510M, D&A $1,393M, interest expense $640M, net loss $626M (all reported). Derived, ours: cash operating costs = revenue - adj. EBITDA ($1,065M; power, data-centre leases, staff, SG&A); other = net loss - (adj. EBITDA - D&A - interest) ($103M; stock comp, tax, other). Cents = item / revenue.

SO WHAT

59% adj. EBITDA margin

54¢ of each $ to depreciation

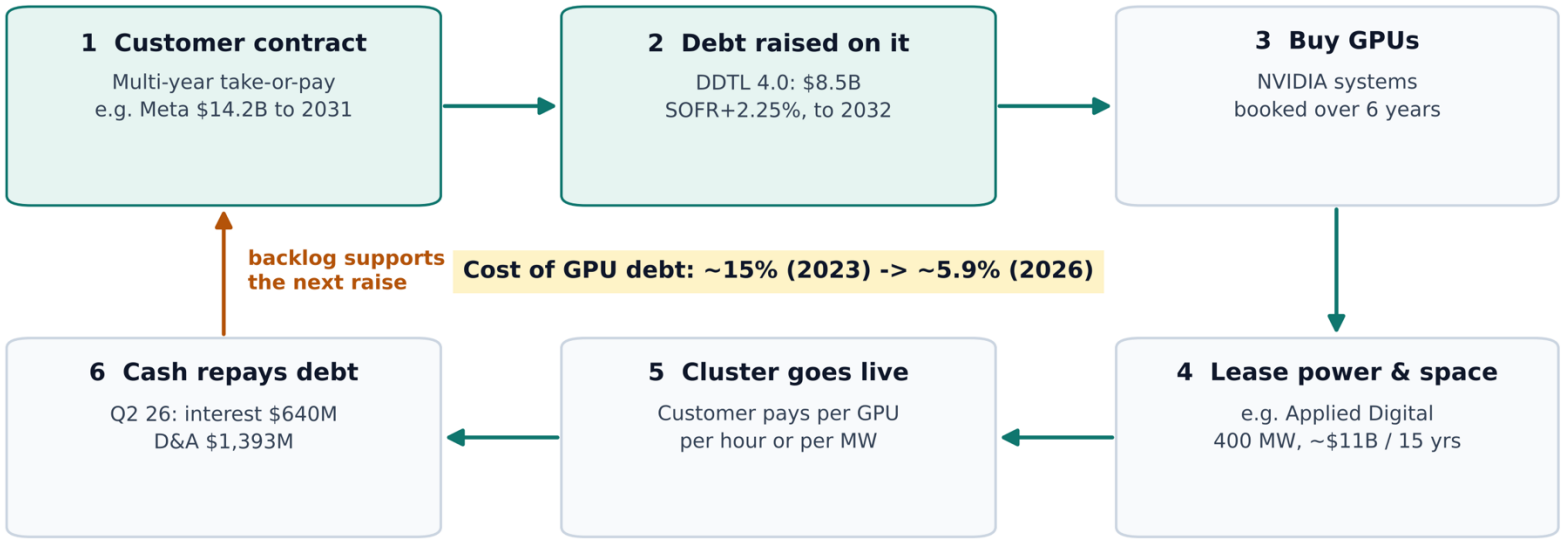
25¢ to interest: $626M net loss

Contracts become collateral for GPU debt

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Exhibit 9: The neocloud financing loop, with CoreWeave examples

Take-or-pay contracts become collateral; the stronger the contract, the cheaper the debt that buys the GPUs



Sources: Meta order $14.2B through Dec 2031 (CoreWeave FY2025 10-K); DDTL 4.0 $8.5B, SOFR+2.25% (~5.9% fixed), matures Mar 2032, backed by a contract with a 'leading AI enterprise' (CoreWeave 8-K, 31 Mar 2026); first GPU-backed facility $2.3B at ~15% (Aug 2023, Forbes, press-reported); Applied Digital leases to CoreWeave 400 MW, ~$11B over ~15 yrs (DCD, press-reported); 6-year GPU book life (S-1, secondary). Q2 2026 interest $640M and D&A $1,393M (release). Schematic, not to scale.

SO WHAT

Debt is secured on customer contracts

GPU loans: ~15% in 2023, ~5.9% in 2026

Works while contracts and GPU values hold

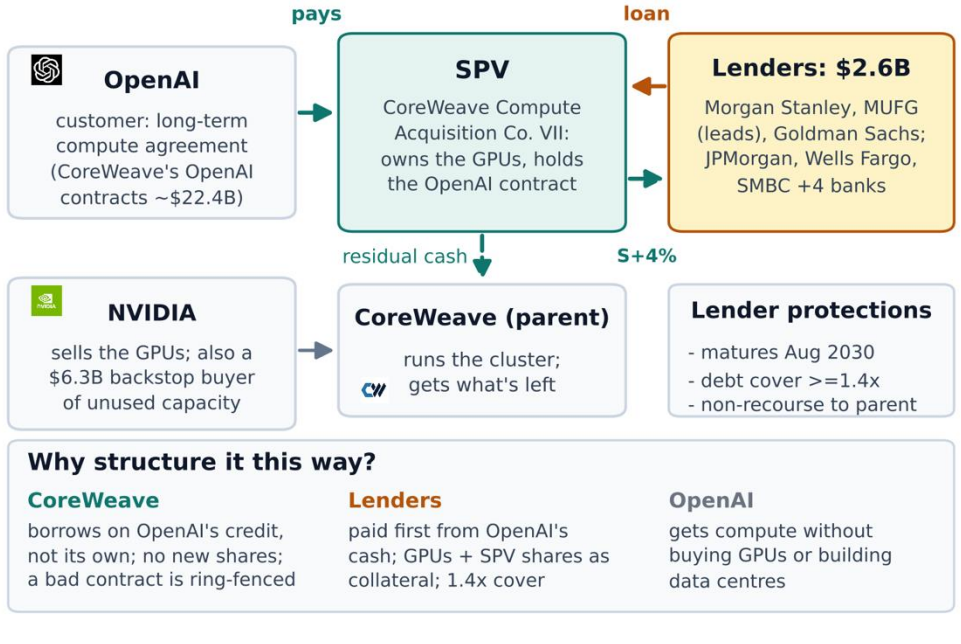
Private credit opened the door to GPU loans

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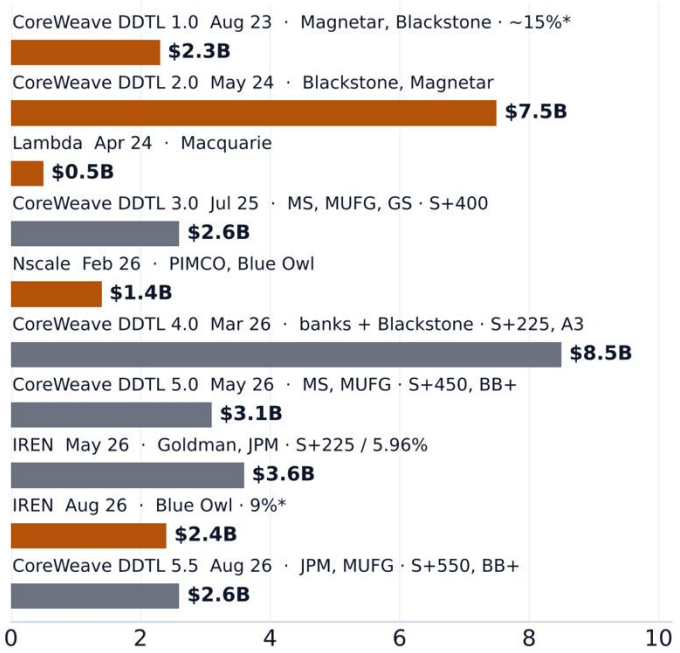
Exhibit 9A: Who lends against GPUs, and how a real deal is wired

Private credit funds wrote the first risky GPU loans; as contracts got stronger, banks and investment-grade buyers piled in and spreads fell

A real deal: CoreWeave's $2.6B loan for its OpenAI contract (Jul 2025)



GPU-backed loans, $B (orange = private credit)



Sources (company unless noted): CoreWeave releases/8-Ks: DDTL 1.0 $2.3B (Aug 2023, Magnetar + Blackstone), DDTL 3.0 $2.6B SOFR+4% (31 Jul 2025, 8-K: borrower CoreWeave Compute Acquisition Co. VII, to serve a long-term OpenAI agreement; leads Morgan Stanley, MUFG, Goldman; also JPMorgan, Wells Fargo, BBVA, Credit Agricole, SMBC, PNC, SocGen; matures Aug 2030; DSCR 1.40x), DDTL 4.0 $8.5B SOFR+2.25% / ~5.9% fixed, Moody's A3, Blackstone anchor (Mar 2026), DDTL 5.0 $3.1B SOFR+4.50% (May 2026); DDTL 5.5 $2.6B SOFR+5.50%, Ba2/BB+, JPMorgan + MUFG (10 Aug 2026, 8-K); DDTL 2.0 $7.5B (May 2024, Blackstone + Magnetar; press: PitchBook, DCD); Nscale $1.4B (Feb 2026, PIMCO, Blue Owl, LuminArx); IREN $2.4B (Aug 2026, Blue Owl; 9% per press) and $3.6B (May 2026, Goldman + JPMorgan; SOFR+2.25% DDTL + 5.96% notes, 8-K); Lambda $500M (Apr 2024, Macquarie). NVIDIA $6.3B residual-capacity backstop to Apr 2032 (filing via CNBC/DCD). DDTL 1.0 ~15% all-in is press-only (Forbes). Schematic, not to scale.

SO WHAT

Private credit led first

OpenAI contract, in an SPV, backs $2.6B

~15% in 2023, S+225 now

The firms financing neocloud GPUs

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Exhibit 9B: The lenders and investors behind neocloud GPUs

Private credit funds took the early GPU-loan risk; big banks joined once Microsoft, Meta and OpenAI signed the contracts behind the loans

	CoreWeave	Crusoe	Lambda	IREN	Nscale	Big Tech SPVs
PRIVATE CREDIT						
Blackstone	DDTL 1.0-2.0 lead; 4.0 anchor					
Blue Owl		Abilene JV $11.6B*		$2.4B lead*	$1.4B co-lead*	Meta Hyperion 80% owner*
Magnetar	DDTL 1.0-2.0 lead; early equity*					
PIMCO	DDTL 1.0*				$1.4B co-lead*	
Brookfield / Upper90		$750M / $225M				
Apollo						xAI GPU SPV (debt)*
Macquarie			$500M (2024)			
Carlyle, BlackRock, Coatue	DDTL 1.0-2.0*					
INVESTMENT BANKS						
Morgan Stanley	DDTL 3.0 $2.6B		$917M (2026)*			
Goldman Sachs	DDTL 3.0			$3.6B, on MSFT contract		
JPMorgan		Abilene loan $7.1B*		$3.6B (with GS)		
MUFG	DDTL 3.0					
EQUITY: PE, SOVEREIGN						
Atreides, Mubadala, Valor		Series F $3.9B co-leads				

Sources: CoreWeave 8-Ks and releases (DDTL 1.0 $2.3B Aug 2023; 2.0 $7.5B May 2024; 3.0 $2.6B Jul 2025; 4.0 $8.5B Mar 2026, Blackstone Credit anchor); lead names for DDTL 1.0-2.0 from PitchBook, DCD and Forbes (press). Crusoe: Brookfield $750M and Upper90 $225M (company); Abilene $11.6B debt+equity in a Blue Owl / Primary Digital JV (DCD) with a JPMorgan-led $7.1B loan (Bloomberg, press); Series F $3.9B co-led by Atreides, Mubadala Capital, Valor (company). Lambda: Macquarie $500M (2024), Morgan Stanley $917M (Aug 2026, press). IREN: Goldman Sachs and JPMorgan $3.6B (May 2026), Blue Owl-led $2.4B (Aug 2026, press). Nscale: $1.4B led by PIMCO, Blue Owl, LuminArx (Feb 2026, press). Meta Hyperion JV 80% Blue Owl (press); xAI GPU SPV with Apollo debt (press). * = press-reported.

SO WHAT

Private credit took the first risk

Banks joined once Big Tech signed

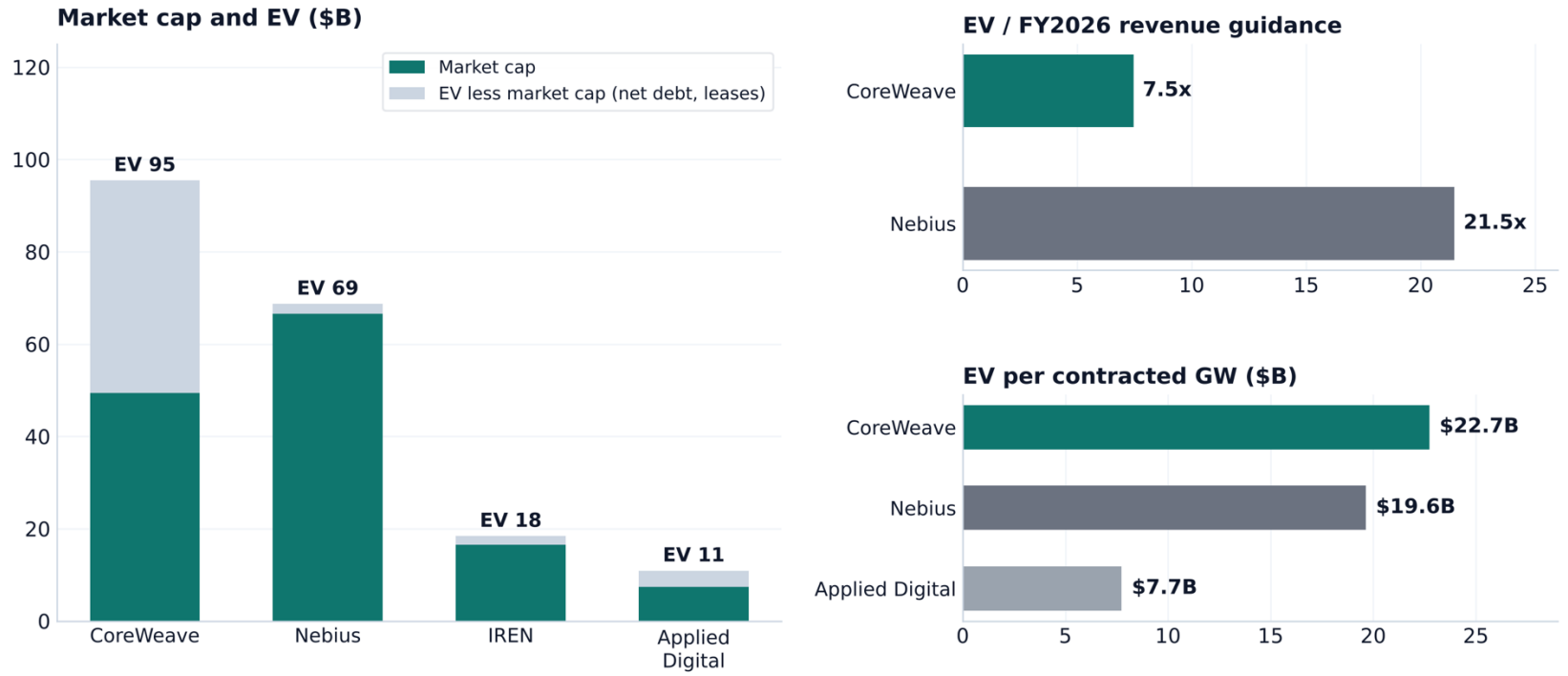
Blue Owl and Blackstone recur most

The market prices them very differently

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Exhibit 10: Public neoclouds: market cap, EV and what the EV buys

Half of CoreWeave's $95B EV is debt; Nebius trades at ~3x CoreWeave's multiple of 2026 revenue



Source: stockanalysis.com market cap and EV, 3-4 Oct 2026 (EV includes lease liabilities). Revenue: company FY2026 guidance midpoints (CoreWeave $12.4-13.2B, Aug 2026; Nebius $3.0-3.4B). Contracted power: CoreWeave 4.2 GW (Aug 2026 call), Nebius >3.5 GW (Q1 2026 letter, lower bound), Applied Digital 1.41 GW (Jul 2026). All ratios = EV / metric, ours. IREN FY guidance not comparable; omitted.

SO WHAT

CoreWeave EV $95B, about half debt

Nebius 21.5x 2026 sales vs CoreWeave 7.5x

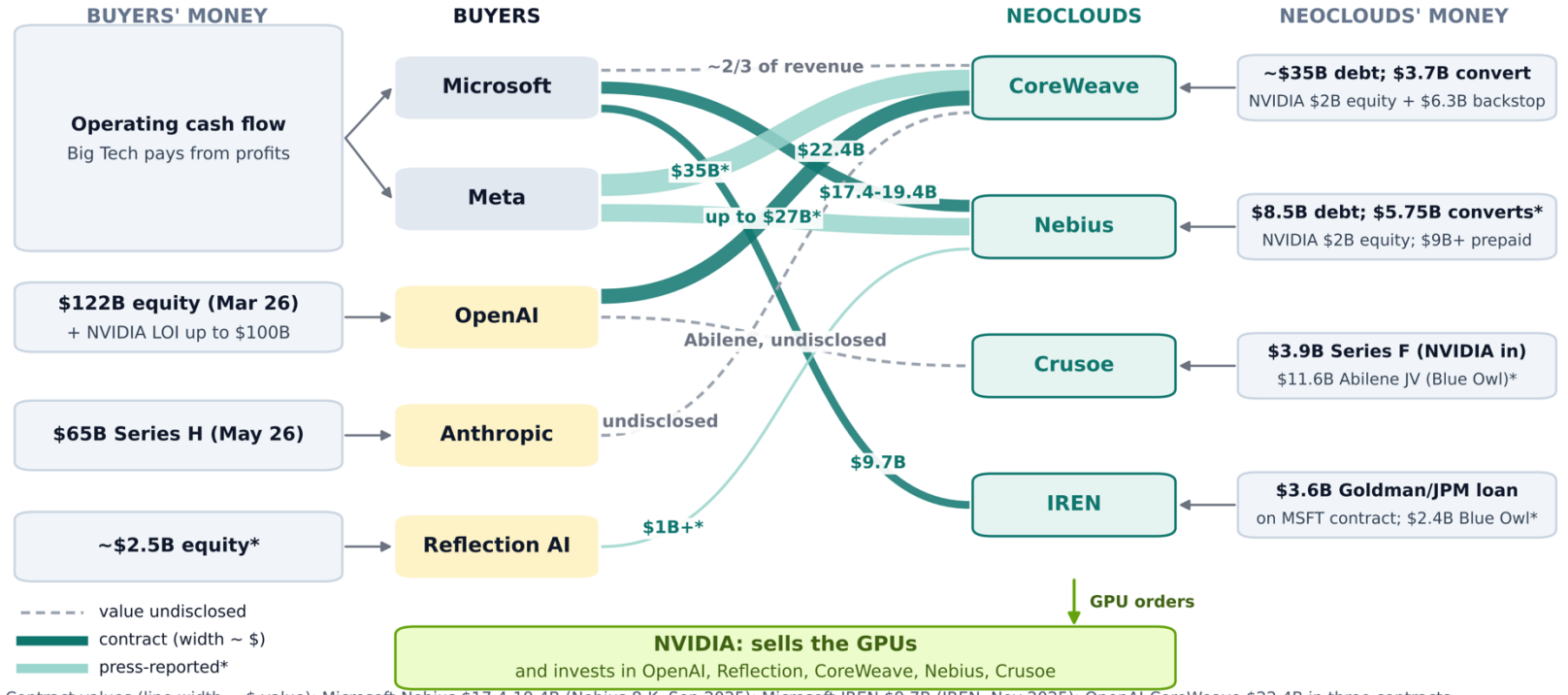
12 months: CoreWeave −35%, Nebius +110%

Who pays neoclouds, and with what money

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Exhibit 10B: Follow the money: who pays neoclouds, and with what

Buyers fund contracts from cash flow or equity; neoclouds turn those contracts into debt; most of the money ends up buying NVIDIA GPUs



Contract values (line width ~ $ value): Microsoft-Nebius $17.4-19.4B (Nebius 8-K, Sep 2025); Microsoft-IREN $9.7B (IREN, Nov 2025); OpenAI-CoreWeave $22.4B in three contracts (CoreWeave releases); Meta-CoreWeave $14.2B (FY2025 10-K) plus $21B expansion (press, Apr 2026); Meta-Nebius up to $27B (press); Reflection-Nebius $1B+ (Bloomberg). Microsoft share of CoreWeave revenue 62% (2024) and ~67% (2025). Anthropic-CoreWeave and OpenAI/Oracle-Crusoe values undisclosed (dashed). Buyer funding: OpenAI $122B round (Mar 2026) and NVIDIA LOI up to $100B (Sep 2025); Anthropic $65B Series H (May 2026); Reflection ~$2.5B (press). Provider funding: company releases and 8-Ks; Nebius converts (DCD), Abilene JV (DCD), IREN Blue Owl loan (press). * = press-reported.

SO WHAT

Big Tech pays from profits; labs from equity

Contracts become collateral for loans

NVIDIA funds both ends, then sells the GPUs

SECTION 3 OF 4

Neolabs

New labs that raise billions before a product

COMING UP

How a neolab works 25–26

Valuations 27

Inside TML, AMI, Periodic, Prime, Reflection 28–37

Beam: Reflection's new open model 38

How each neolab makes money 39

The prize, the money, NVIDIA's loop 40–43

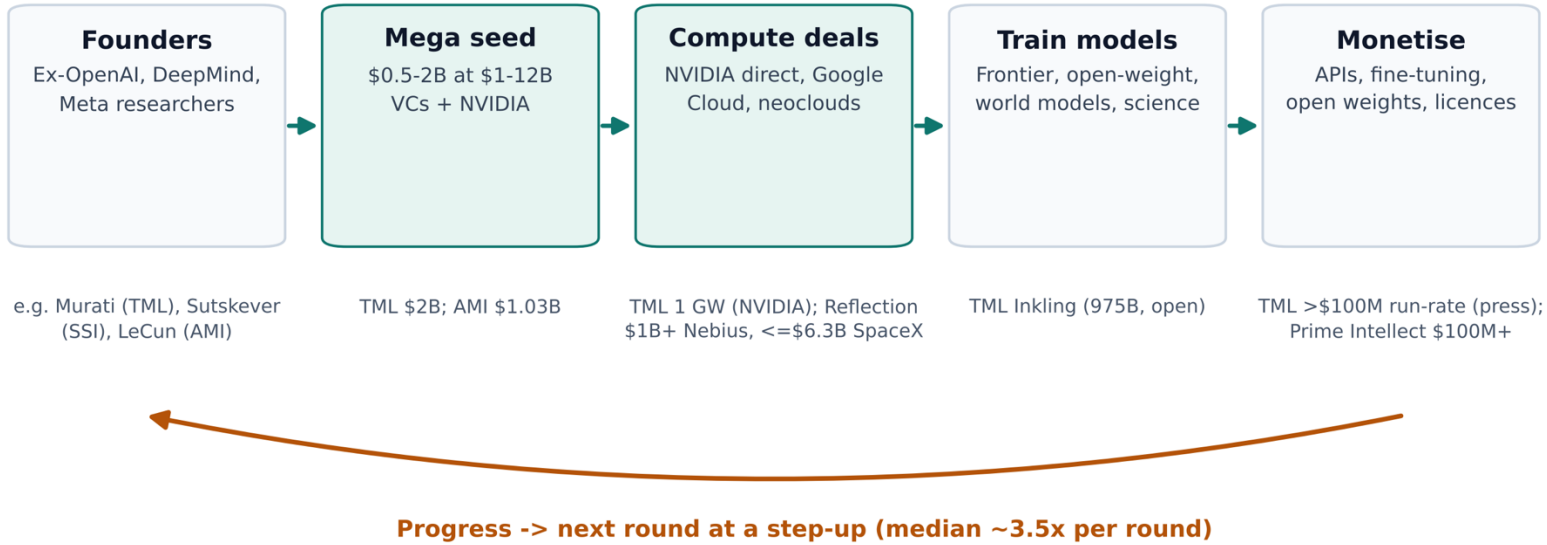
AMD buys World Labs 44

A neolab sells a research bet

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Exhibit 11: How a neolab works: talent and capital in, compute and models out

A neolab sells a research bet: star founders raise billions pre-product, spend most of it on compute, then step up again



Sources: Thinking Machines $2B seed at $12B post (Jul 2025) and NVIDIA 1 GW Vera Rubin deal (NVIDIA blog, Mar 2026); AMI Labs $1.03B seed (TechCrunch, Mar 2026); Reflection-Nebius $1B+ compute (TechCrunch, Jul 2026) and SpaceX Colossus 2 up to $6.3B (CNBC, Jun 2026); SSI moving from Google TPUs to NVIDIA (Jul 2026); Thinking Machines Tinker API and Inkling open-weight model (company); revenue: Thinking Machines >$100M run-rate (The Information, press-reported), Prime Intellect $100M+ (SiliconANGLE); median 3.53x valuation step-up per round across 23 neolabs (J. Seidel analysis, Aug 2026, unaudited). Schematic.

SO WHAT

Star founders raise billions pre-product

Most of it goes on compute

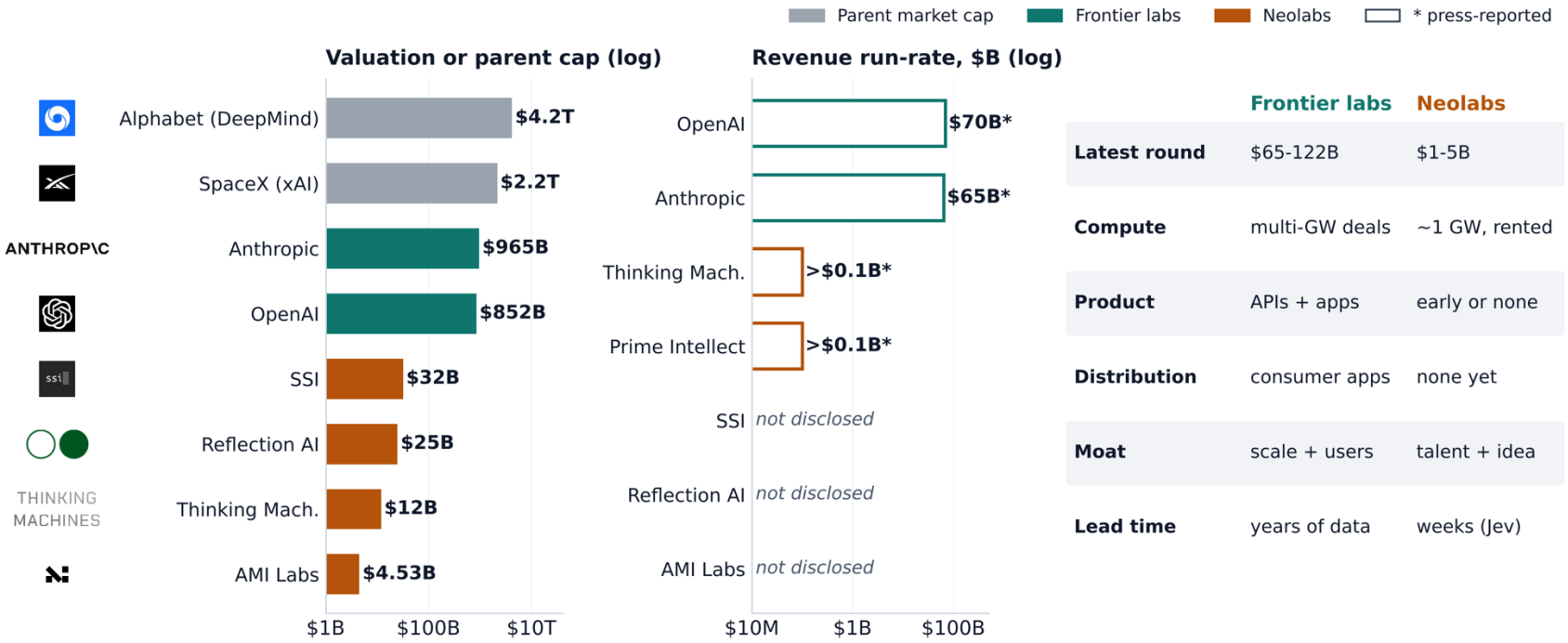
Progress buys a ~3.5x step-up

Frontier labs sell; neolabs still research

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Exhibit 11A: Frontier labs vs neolabs: valuation, revenue and resources

Frontier labs are valued at ~$0.9-1T on ~$65-70B of run-rate revenue; neolabs are valued at $5-32B mostly on talent, with little revenue yet



	Frontier labs	Neolabs
Latest round	$65-122B	$1-5B
Compute	multi-GW deals	~1 GW, rented
Product	APIs + apps	early or none
Distribution	consumer apps	none yet
Moat	scale + users	talent + idea
Lead time	years of data	weeks (Jev)

Log scales. Valuation (post-money unless noted): Anthropic $965B (Series H, May 2026, company); OpenAI $852B (Mar 2026, company); SSI $32B (Apr 2025, press); Reflection $25B pre-money (CEO on CNBC, Apr 2026); Thinking Machines $12B (Jul 2025, press; $40B+ round in talks); AMI Labs $4.53B (Mar 2026, press). Grey = parent market cap, not a round: Alphabet ~$4.2T and SpaceX (incl. xAI) ~$2.16T at 2 Oct 2026 close (stockanalysis.com, aggregator). Revenue run-rate: OpenAI ~$70B (Axios, Sep 2026), Anthropic $65B (Bloomberg, Jul 2026), Thinking Machines >$100M (The Information via press, Sep 2026; another report says several hundred million), Prime Intellect $100M+ (investor Intel Capital). SSI, Reflection and AMI disclose no revenue. Run-rate is annualised, not GAAP. Comparison table: ARC Research summary of facts cited elsewhere in this deck.

SO WHAT

Valuations 30-210x apart

$65-70B revenue vs ~$0.1B

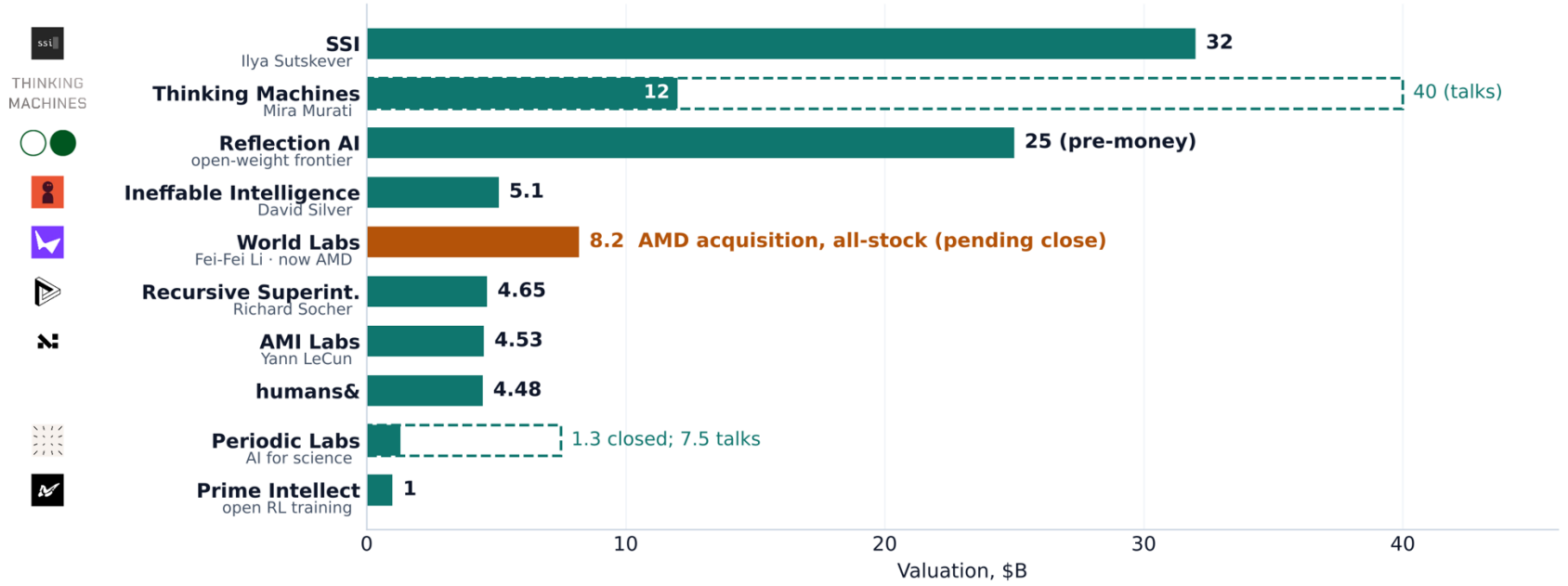
Neolabs are priced on talent

Billion-dollar valuations before a product

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Exhibit 12: Neolab valuations: last closed round vs rounds in talks ($B)

Ten labs worth ~$90B on closed rounds; World Labs is the first exit, to AMD for $8.2B in stock



Source (all press-reported unless noted): SSI $32B (Apr 2025, TechCrunch); Thinking Machines $12B post (Jul 2025), $40B+ talks (The Information via PYMNTS, Sep 2026); Reflection $25B pre-money, round closed per CEO on CNBC (Apr 2026; ~$2.5B size press-reported; previous $8B, Oct 2025); Ineffable $5.1B (Tech.eu, Apr 2026); World Labs: AMD to acquire for ~$8.2B all-stock (AMD press release, 28 Sep 2026; close expected end-2026); Recursive $4.65B post (TNW, May 2026); AMI Labs $4.53B post (TechCrunch, Mar 2026); humans& $4.48B (Jan 2026); Periodic $1.3B (Sep 2025), $7.5B talks (Forbes, May 2026); Prime Intellect $1B (SiliconANGLE, Jul 2026). Dashed = in talks / not closed; amber = acquisition. Combined closed total = sum of teal bars, ours.

SO WHAT

SSI: $32B, no public product

World Labs → AMD: $8.2B, first exit

~$90B across ten labs, closed rounds

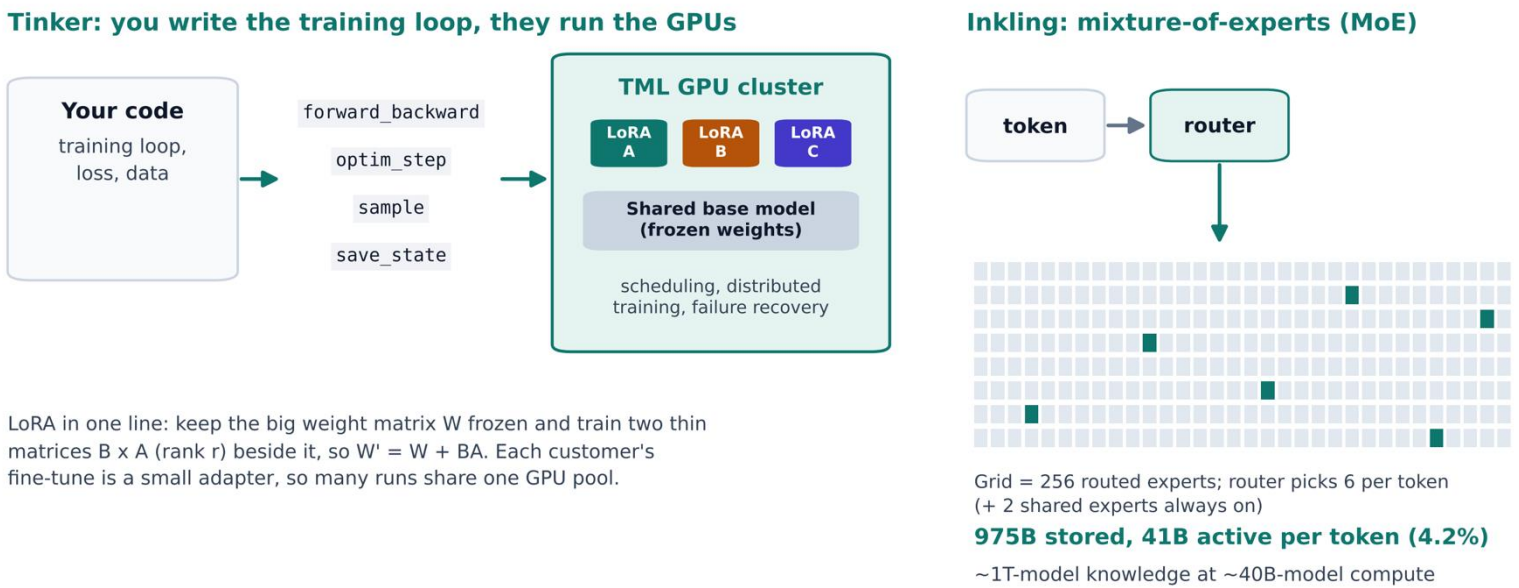
Thinking Machines rents out training

THINKING MACHINES

ARC RESEARCH | AI UPDATE | INSIDE THE NEOLABS

Exhibit 12A: Thinking Machines: Tinker fine-tuning API and the Inkling model

Sells training as a service on shared GPUs (LoRA) and ships open-weight models that only wake ~4% of their weights per token



Sources (company): thinkingmachines.ai Tinker page and 'Announcing Tinker' (1 Oct 2025): API primitives, LoRA, usage-based per-token pricing; 'LoRA Without Regret' blog (29 Sep 2025): rank 256-512 LoRA matches full fine-tuning on instruction and reasoning data, best LoRA learning rate ~10x full fine-tuning; Inkling model card and release (15 Jul 2026): 975B total / 41B active, 256 routed experts, 6 active + 2 shared, 1M context, Apache 2.0, text/image/audio in. 41 / 975 = 4.2%, ours. Schematic, not to scale.

FOUNDERS

Mira Murati
CEO · ex-OpenAI CTO

John Schulman
Chief scientist · OpenAI co-founder

Lilian Weng
ex-OpenAI safety research lead

KEY FACTS

Funding
$2B seed at $12B; $1B at $40B+ in talks*

Compute
Google Cloud GB300; NVIDIA 1 GW Rubin*

Customers
Tinker users; >$100M run-rate*

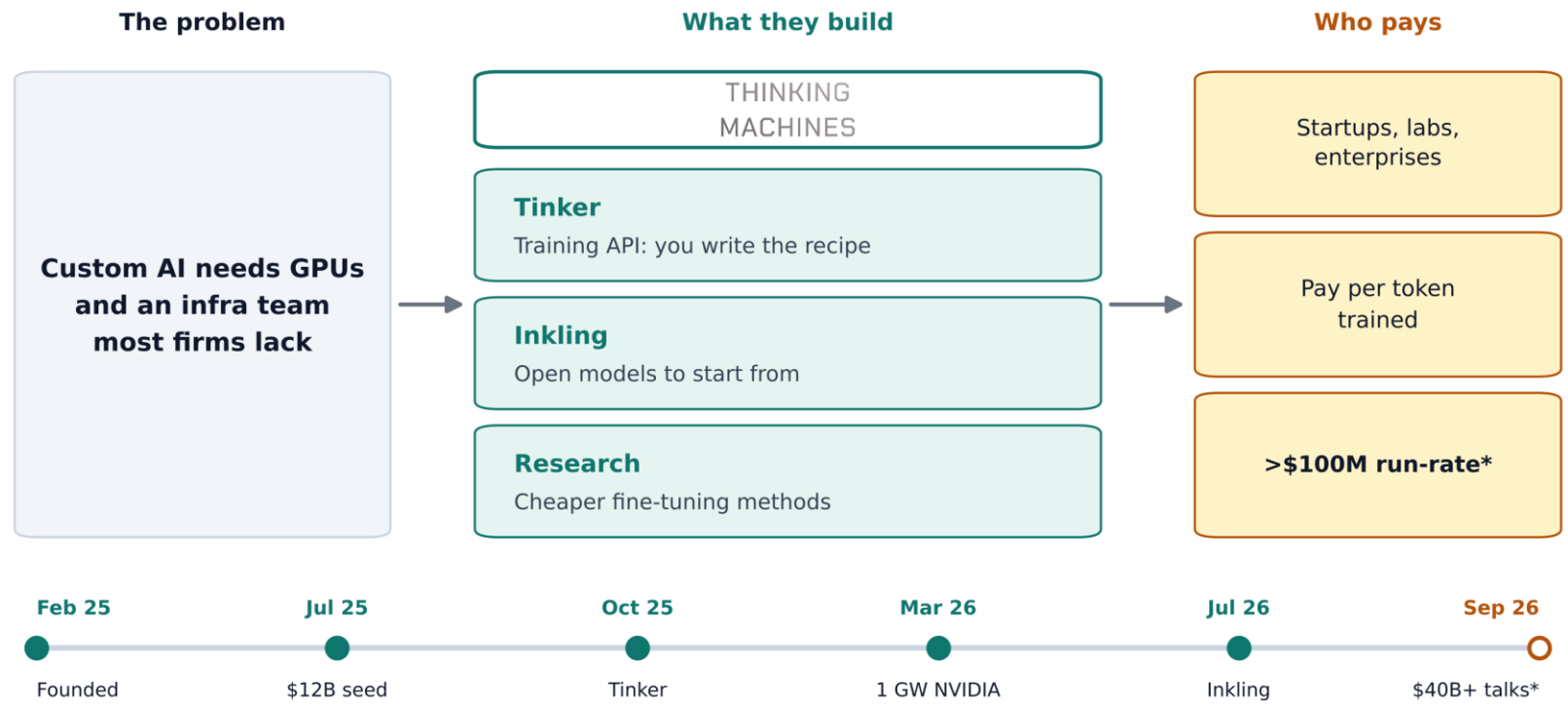
What Thinking Machines does

THINKING MACHINES

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Exhibit 12A2: What Thinking Machines does

You bring the training recipe; it runs the GPUs and hands back your model



Sources: thinkingmachines.ai; Reuters (Jul 2025); WSJ/Reuters (Mar 2026); TechCrunch (Sep 2026). Hollow = press-reported, not closed. * = press-reported.

>$100M*
run-rate

975B
Inkling parameters

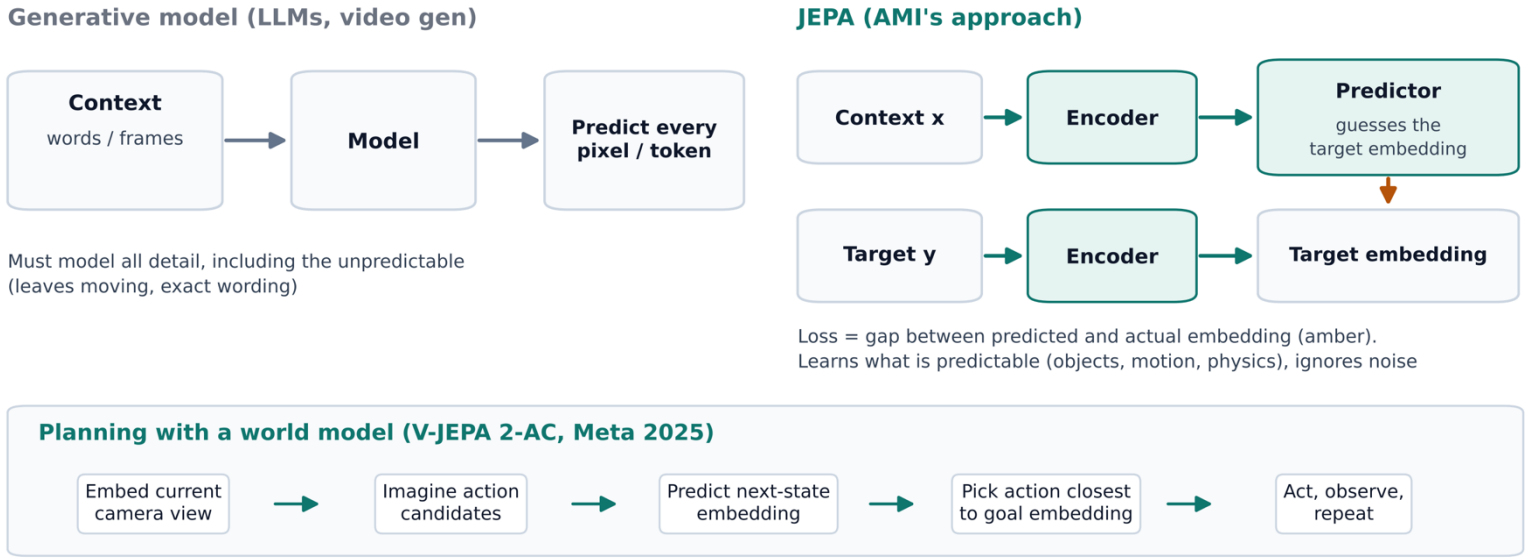
$12B
last closed round

AMI bets against next-token prediction

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Exhibit 12B: AMI Labs: world models that predict meaning, not pixels (JEPA)

Instead of generating the next word or pixel, a JEPA predicts an abstract summary of what happens next, then plans actions with it



Sources: Assran et al., I-JEPA, arXiv 2301.08243 (2023); Meta, V-JEPA 2, arXiv 2506.09985 and Meta AI blog (11 Jun 2025): 1.2B parameters pretrained on 1M+ hours of video; action-conditioned V-JEPA 2-AC trained on <62 hours of unlabelled robot video, zero-shot planning on Franka arms, 65-80% pick-and-place success on new objects (Meta). AMI's approach and funding: press (Futurum, Mar 2026); AMI has published no paper of its own yet. Healthcare partner Nabla per HLTH (Mar 2026). Schematic, not to scale.

FOUNDERS

Yann LeCun
Exec. chairman · Turing Award, ex-Meta chief AI scientist

Alexandre LeBrun
CEO · founded Nabla

KEY FACTS

Funding
$1.03B seed at $3.5B pre (Mar 26)*

Compute
Not disclosed

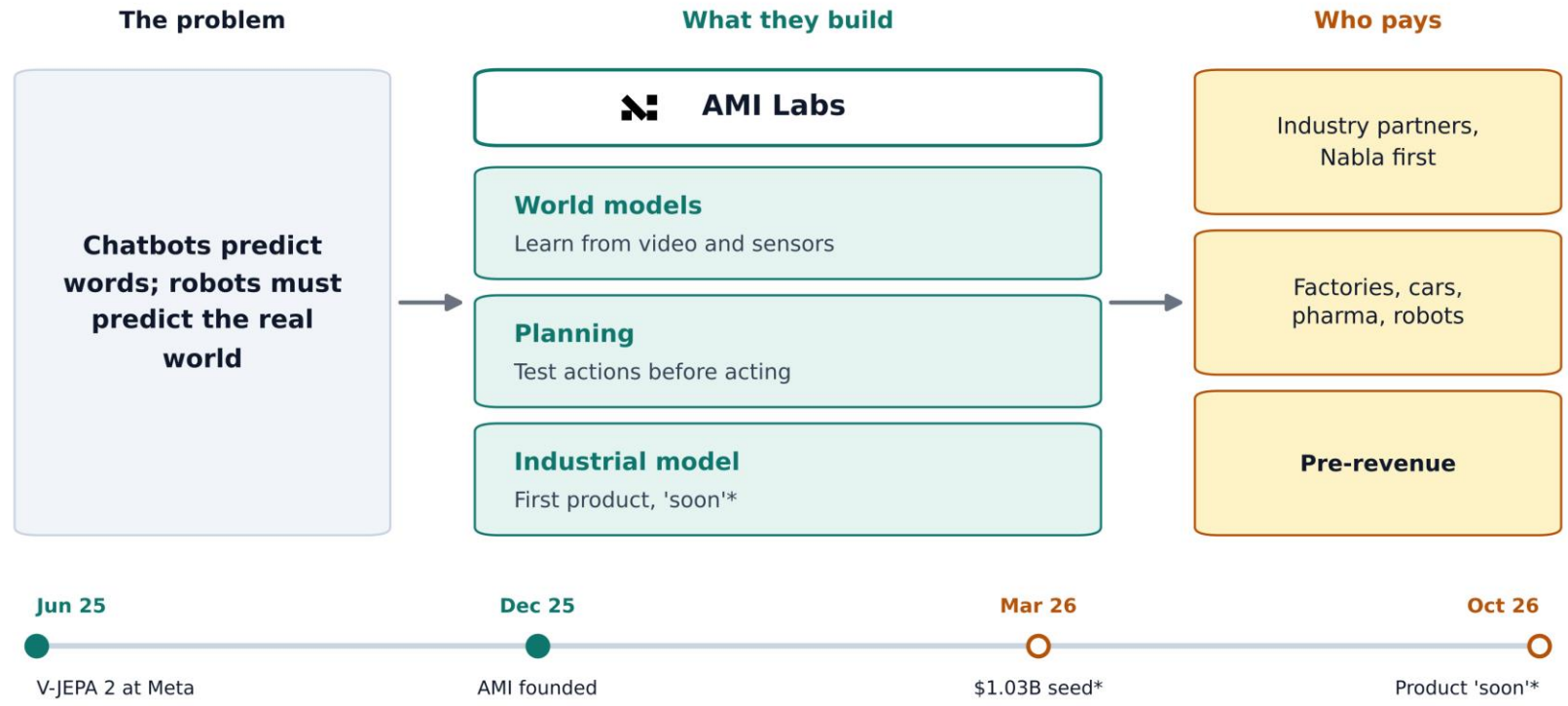
Customers
Pre-revenue; Nabla health partner

What AMI Labs does

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Exhibit 12B2: What AMI Labs does

It teaches AI how the physical world behaves, so machines can plan before they act



Sources: amilabs.xyz; Meta V-JEPA 2 (Jun 2025); TechCrunch (Mar 2026); Fortune interview with Yann LeCun (1 Oct 2026). * = press-reported.

$1.03B*
seed raised

~60*
staff

0
products shipped

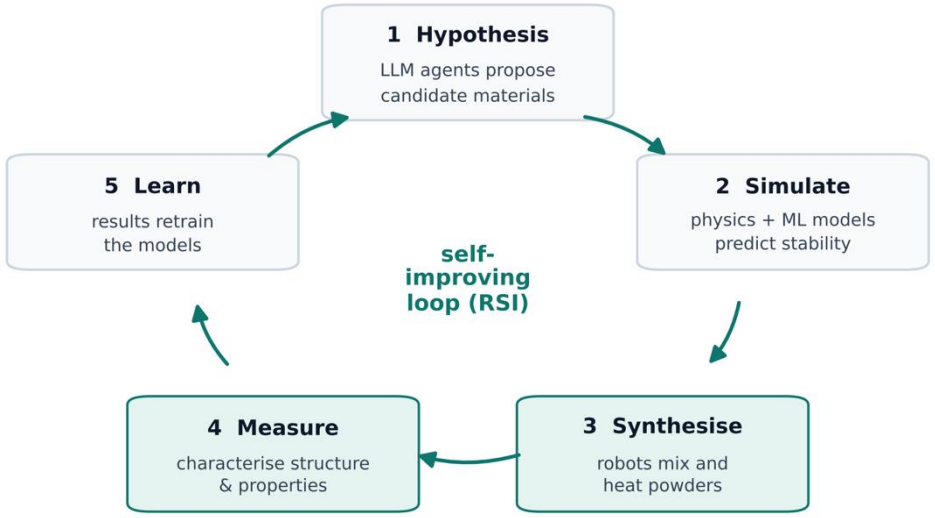
Periodic is building self-improving AI for science

periodic labs

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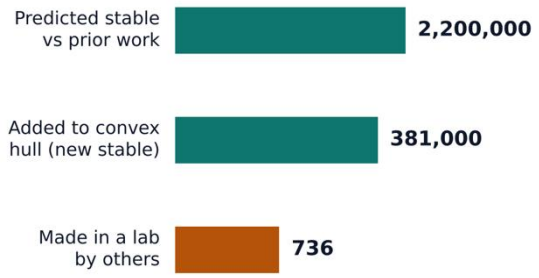
Exhibit 12C: Periodic Labs: recursive self-improvement (RSI) for science

Each robot experiment retrains the model that picks the next one: a self-improving loop on data the internet doesn't have



Why the founders think it works: GNoME

DeepMind's 2023 materials model (Cubuk co-author)



Bars on a log scale. The gap between predicted and made is the bottleneck Periodic's robots target.

Target: superconductors above ~135 K

Sources: periodic.com (company); Cognitive Revolution podcast with founders Liam Fedus and Ekin Dogus Cubuk (press): powder synthesis first, goal of superconductors above the ~135 K ambient-pressure record, ~30 staff; Maginative (30 Sep 2025): $300M seed led by a16z; Merchant et al., 'Scaling deep learning for materials discovery' (GNoME), Nature, 29 Nov 2023: 2.2M structures predicted stable vs prior work, 381,000 added to the convex hull, 736 independently realised by experiment (Cubuk co-author). Loop: Contrary Research (Jun 2026): 'nature is our RL environment'; experiments as the reward signal. RSI here = the loop improves the science model, not general AI. Schematic, not to scale.

FOUNDERS

Liam Fedus
ex-OpenAI · helped create ChatGPT

Ekin Dogus Cubuk
ex-Google DeepMind · GNoME materials AI

KEY FACTS

Funding
$300M seed at $1.3B; $500M+ at $7.5B in talks*

Compute
Not disclosed

Customers
Early chip-industry customers, unnamed*

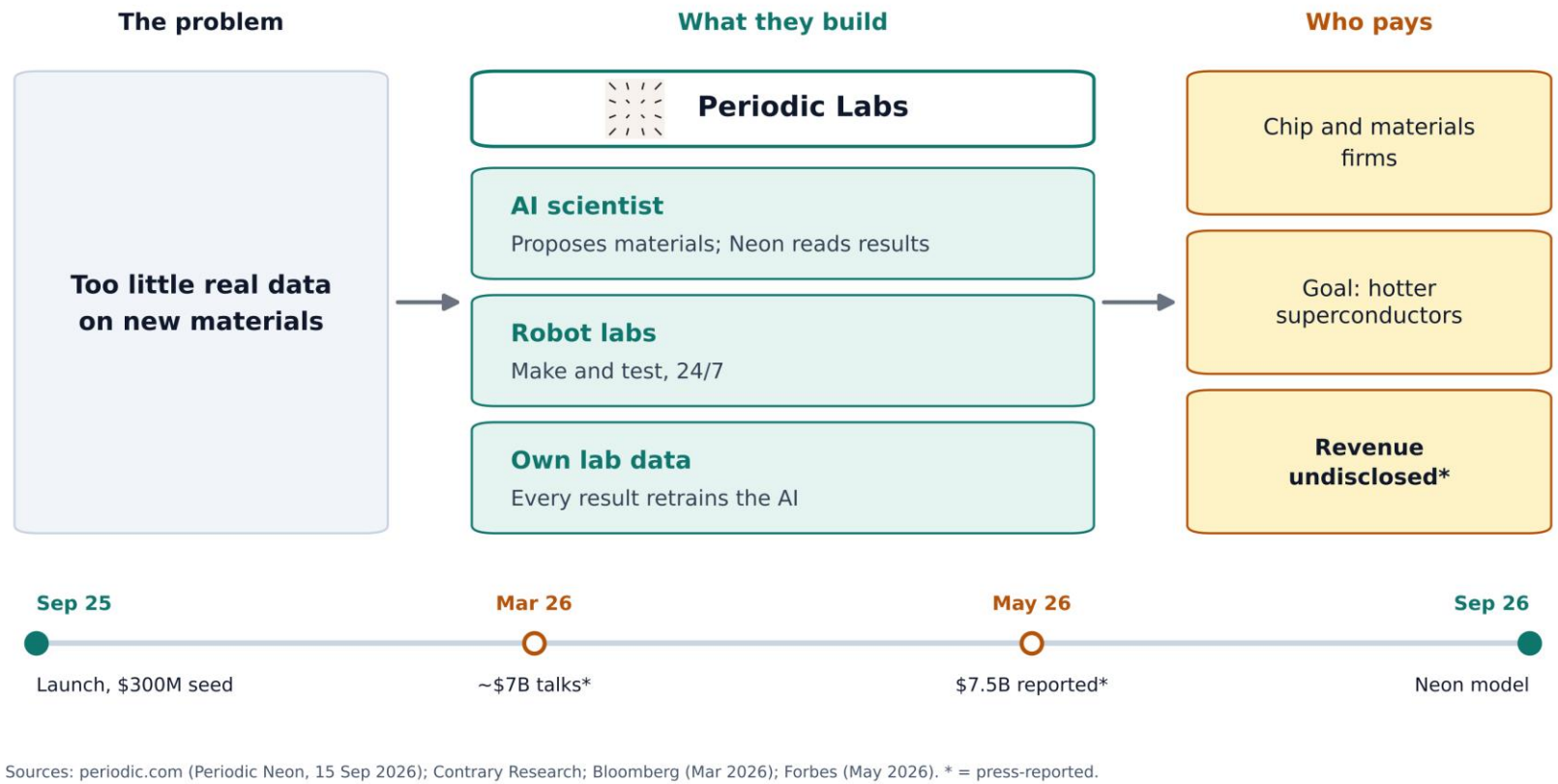
What Periodic Labs does

periodic labs

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Exhibit 12C2: What Periodic Labs does

AI proposes new materials, robots make and test them, results make the AI smarter



Sources: periodic.com (Periodic Neon, 15 Sep 2026); Contrary Research; Bloomberg (Mar 2026); Forbes (May 2026). * = press-reported.

1T
Neon parameters

$300M
seed

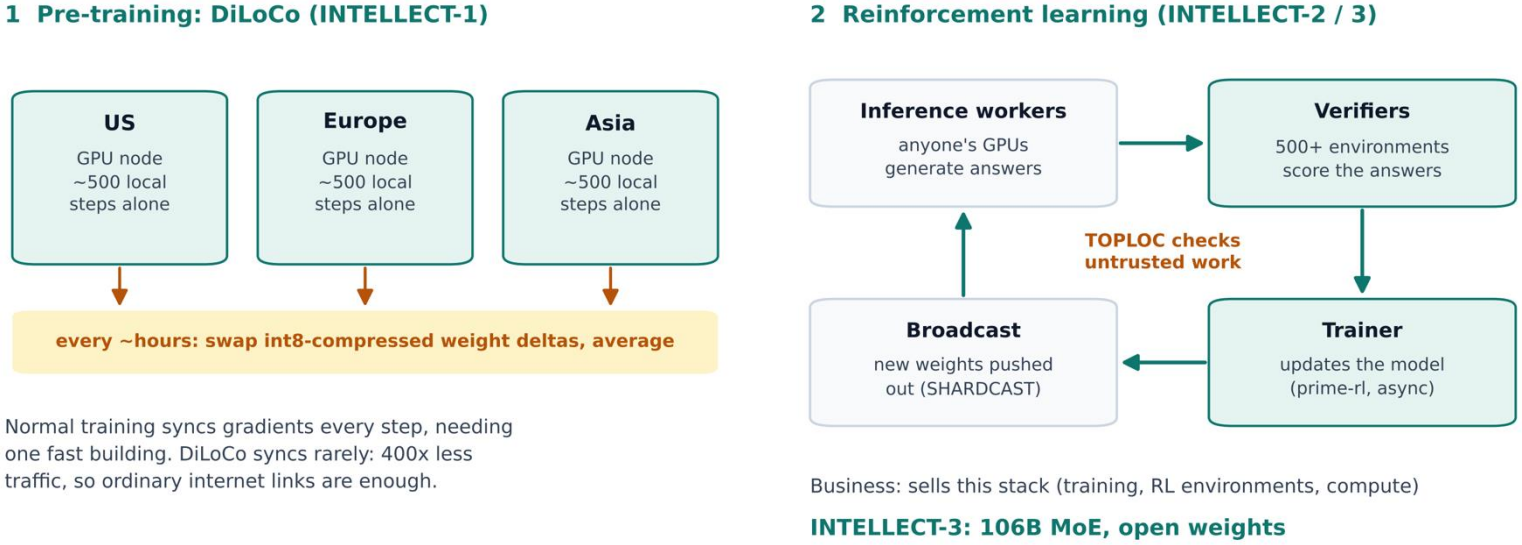
135 K
record to beat

Prime Intellect trains across scattered GPUs

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Exhibit 12D: Prime Intellect: training models across many data centres at once

Low-communication training lets scattered, rented GPUs act as one cluster; then RL runs on untrusted volunteer inference



Sources: Prime Intellect, INTELLECT-1 release (29 Nov 2024) and arXiv 2412.01152: 10B params, 1T tokens, 42 days, up to 112 H100s on 3 continents, 30 compute providers, DiLoCo + int8 all-reduce, 400x less communication than data-parallel, MFU 36-41%; Douillard et al., DiLoCo, arXiv 2311.08105 (Google DeepMind); INTELLECT-2 (11 May 2025, arXiv 2505.07291): 32B, async RL, TOPLOC, SHARDCAST; INTELLECT-3 (26 Nov 2025): 106B MoE on GLM-4.5-Air, 512 H200s, prime-rl, verifiers, 500+ environments; Series A $130M (Intel Capital post, Jul 2026). Schematic, not to scale.

FOUNDERS

Vincent Weisser
Co-founder & CEO

Johannes Hagemann
Co-founder

John Schulman
Angel investor · OpenAI co-founder

KEY FACTS

Funding
$130M Series A (Jul 26), ~$1B val.*

Compute
Aggregates 12+ clouds; Nebius preferred

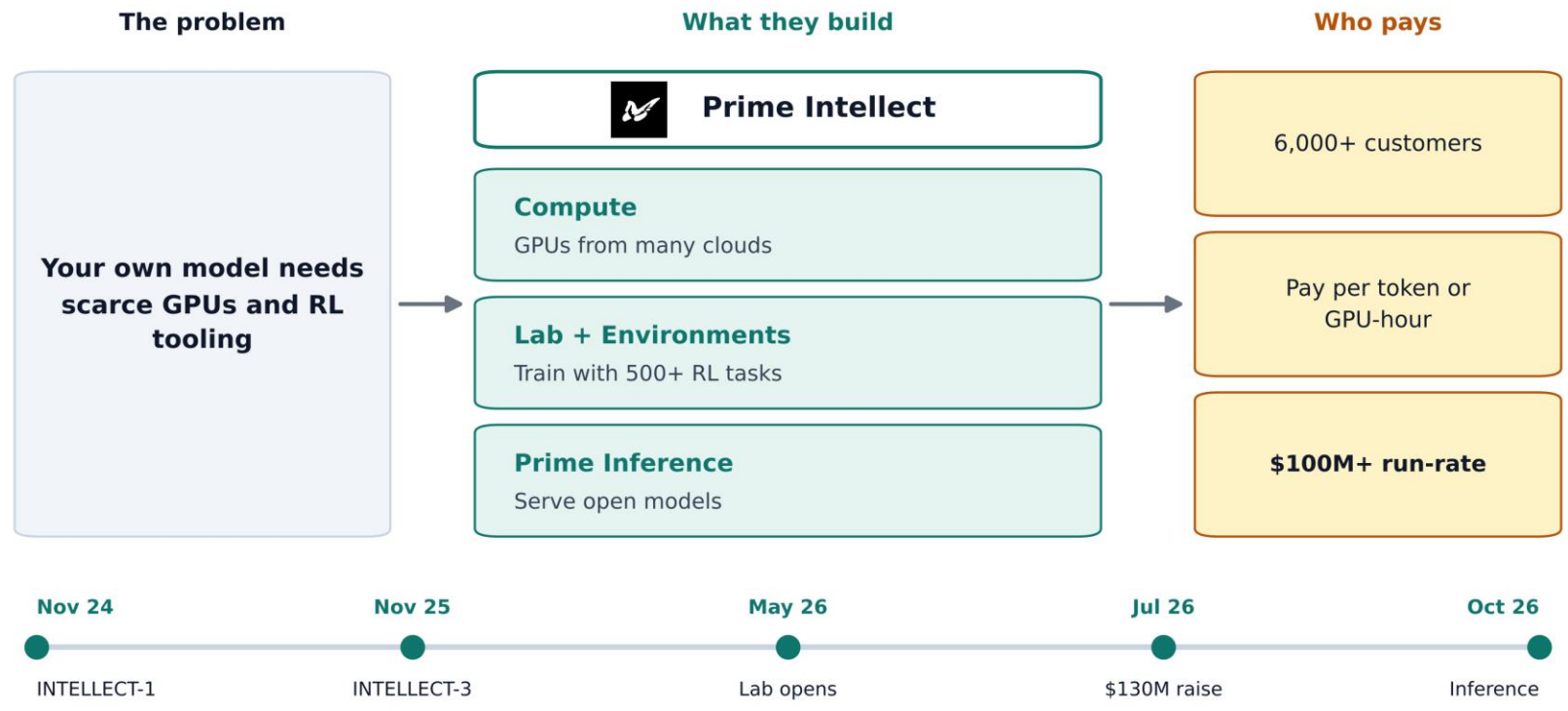
Customers
6,000+, $100M+ ARR: Ramp, NVIDIA, Zapier

What Prime Intellect does

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Exhibit 12D2: What Prime Intellect does

One place to rent GPUs, train your own open model and serve it



Sources: primeintellect.ai (Lab, 7 May 2026; Prime Inference, 2 Oct 2026); Intel Capital (Jul 2026); SiliconANGLE. * = press-reported.

6,000+
customers

$100M+
revenue run-rate

500+
RL practice tasks

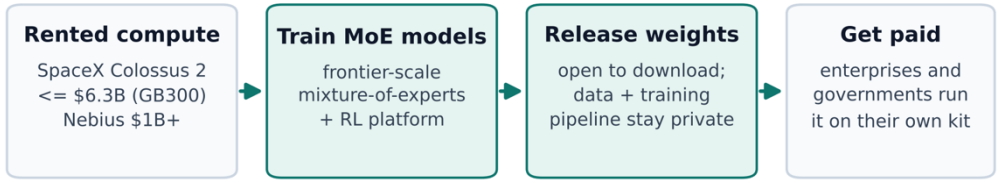
Reflection gives away weights, sells deployment

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Exhibit 12E: Reflection AI: an American open-weight frontier lab

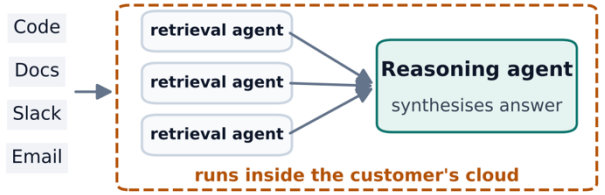
Trains large mixture-of-experts models on rented compute, gives away the weights, and sells deployment to enterprises and governments

The business model

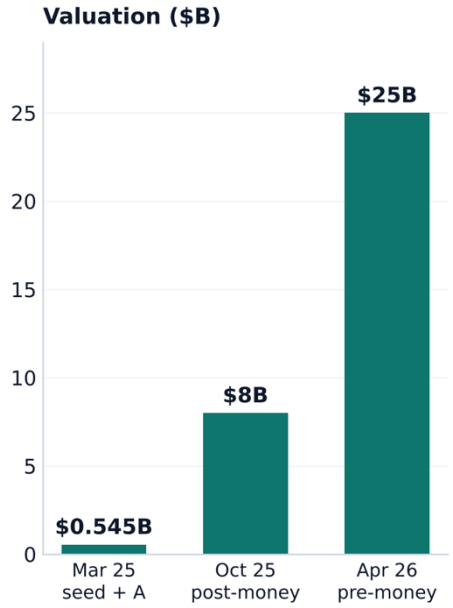


Dell on-prem AI Factory
DOE Genesis Mission
Korea 250 MW DC (MoU)

Asimov: code-comprehension agent (Jul 2025)



82% of answers preferred vs
63% for Claude Sonnet 4
(Reflection's own survey)



Sources: Reflection blog 'Frontier open intelligence' (Oct 2025, company): $2B raised; platform to train 'massive Mixture-of-Experts models at frontier scale'; weights to be released, data and pipelines kept (TechCrunch). Asimov (SiliconANGLE, Jul 2025): retrieval agents + reasoning agent inside the customer's cloud; preferred on 82% of prompts vs 63% for Claude Sonnet 4 in Reflection's own developer survey. Compute: SpaceX Colossus 2 up to $6.3B, Jul 2026-2029, ~$150M/month, GB300 (Qz/CNBC); Nebius $1B+ (Bloomberg). Customers: Dell on-prem AI Factory (May 2026), DOE Genesis Mission (Axios, May 2026), Shinsegae 250 MW Korea MoU (DCD, Mar 2026). Valuations: $545M (Mar 2025, Bloomberg via Wikipedia), $8B post (Oct 2025, TechCrunch), $25B pre-money (CEO on CNBC, Apr 2026). All press-reported unless noted. Schematic.

Reflection

FOUNDERS

Misha Laskin
CEO · led Gemini reward modelling

Ioannis Antonoglou
CTO · AlphaGo, AlphaZero, MuZero

KEY FACTS

Funding
Closed at $25B pre (Apr 26), ~$2.5B*

Compute
SpaceX Colossus 2 to $6.3B; Nebius $1B+*

Customers
Dell, US DOE Genesis; Shinsegae MoU

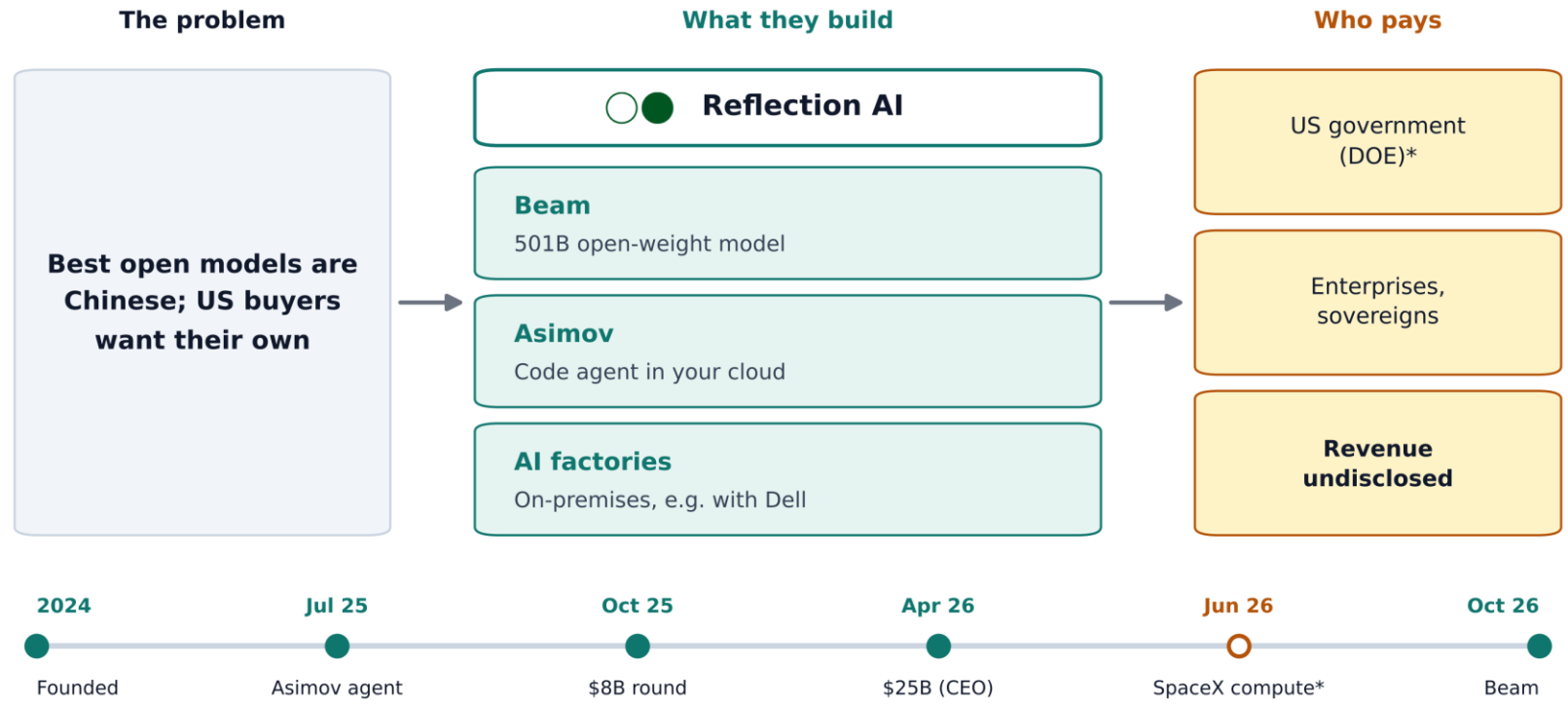
What Reflection AI does

Reflection

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Exhibit 12E2: What Reflection AI does

A US open model anyone can download, with paid help to run it privately



Sources: reflection.ai (Beam, 5 Oct 2026); TechCrunch; CNBC (Apr 2026); Bloomberg (compute*); Axios (DOE*). * = press-reported.

501B
Beam parameters

$25B
valuation (pre-money)

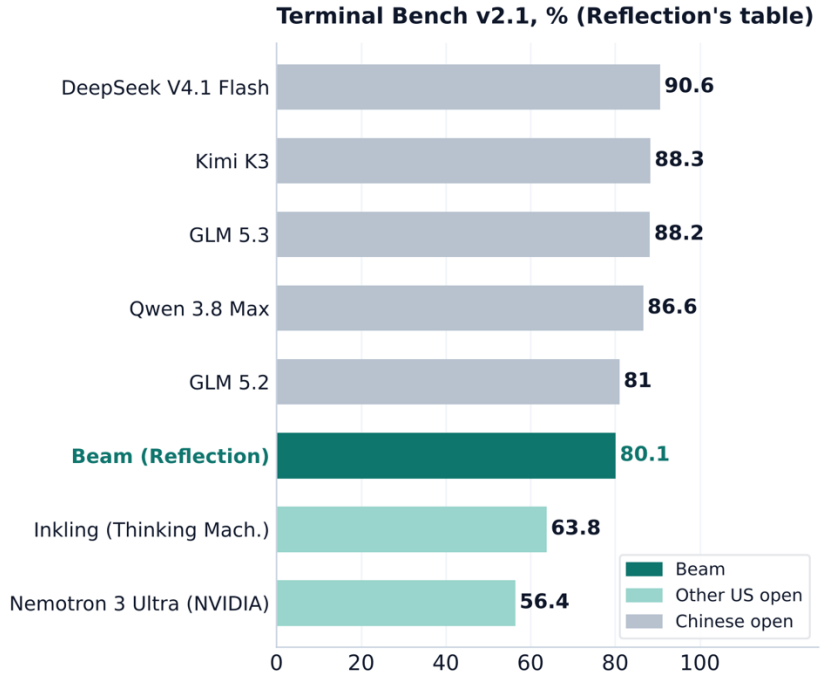
$7.3B*
rented compute

Reflection launches Beam, a US open model

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Exhibit 12F: Reflection's Beam: a US open-weight model, launched 5 Oct 2026

Beam leads the US open models on Reflection's own tests but trails China's newest; NVIDIA backs it with money and GPUs



Beam at a glance

Size	501B params, 23B active (MoE)
Context	256K, extendable to 1M
Training	23.8T tokens
Pretrain	6,144 NVIDIA GB300, <4 weeks
RL	10.5K GB300, 4 weeks
Weights	Apache 2.0, later in October

Why NVIDIA backs US open models

Investor	among leads of Reflection's $2B round (Oct 2025)*
Compute	Beam trained on GB300; Nemotron Coalition offers DGX Cloud to Reflection, TML (Mar 2026)*
Demand	every open model, run anywhere, needs GPUs: 'Free AI should be great for chips' (Huang)*
Policy	US plan wants 'leading open models founded on American values' (Jul 2025)

Benchmarks and specs: Reflection blog 'Introducing Beam' (5 Oct 2026, company); scores are as reported by Reflection, not independently verified; Terminal Bench v2.1 is the only row Reflection reports for all eight models. Weights to be released under Apache 2.0 later in Oct 2026 (company). NVIDIA: among the leads of Reflection's $2B round (Oct 2025, press); Nemotron Coalition with Reflection and Thinking Machines, NVIDIA offering DGX Cloud (16 Mar 2026; DCD, SiliconANGLE, press); Huang 'Free AI should be great for chips' (Aug 2026, press); White House AI Action Plan (Jul 2025, primary): 'We need to ensure America has leading open models founded on American values.' * = press-reported.

SO WHAT

501B open model, 23B active

Tops US open models; trails China's newest

NVIDIA funds it, powers it, wants more

How each neolab plans to make money

Lab	What it sells	Who pays	Revenue today
Thinking Machines	Training API (Tinker) + open models	Developers fine-tuning models	>$100M run-rate*
SSI	Nothing yet: safe superintelligence	—	None
Reflection AI	Free open weights; paid deployment	Enterprises, governments	Not disclosed
AMI Labs	World models licensed to industry	Partners; first Nabla (health)	Pre-revenue
Periodic Labs	AI + robot labs that find materials	Chip and materials firms	Early, undisclosed
Prime Intellect	Compute, RL environments, training	6,000+ firms (Ramp, Zapier)	$100M+ ARR

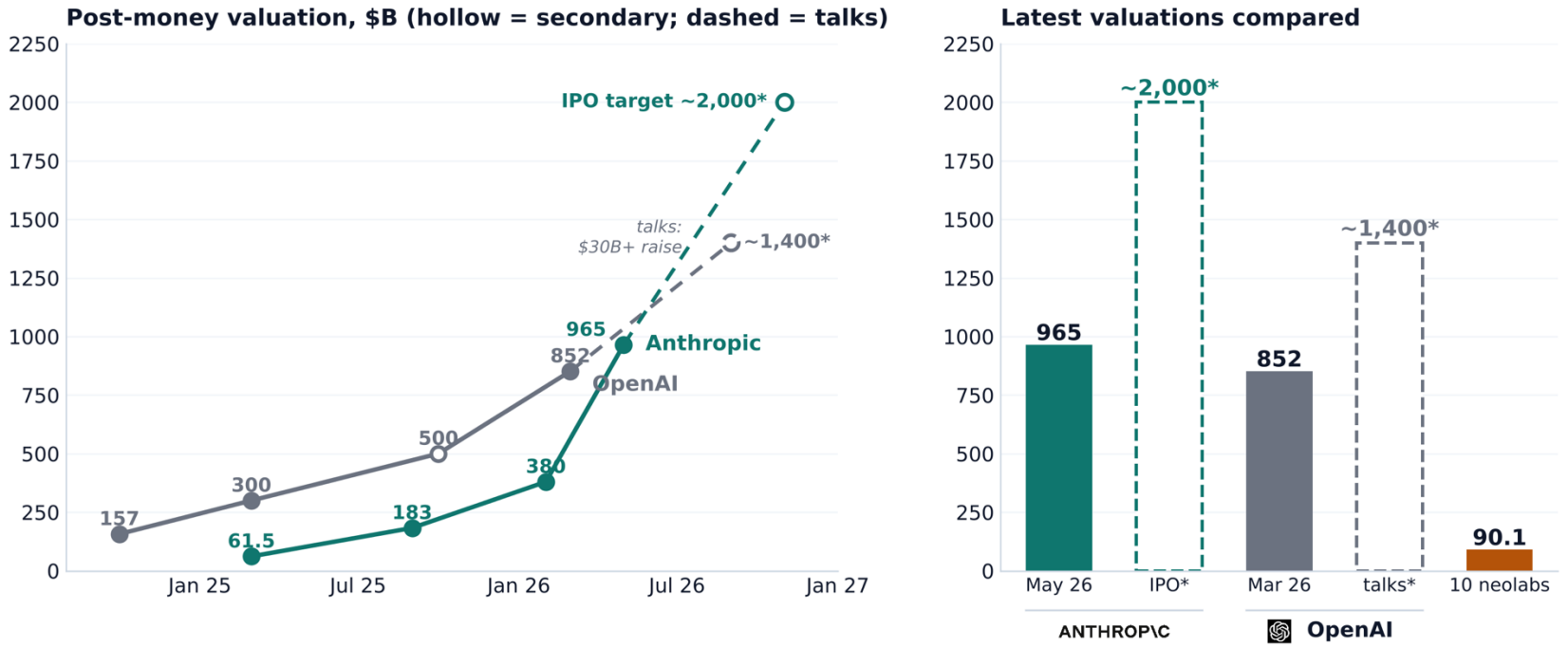
Only two report $100M+ a year; * press-reported

The bet: becoming the next OpenAI or Anthropic

ARC RESEARCH | AI UPDATE | HOW NEOLABS WORK

Exhibit 13: The prize: OpenAI and Anthropic vs the neolab cohort ($B)

Anthropic eyes a ~$2T IPO* after rising 15.7x to $965B; OpenAI is in talks at ~$1.4T*; ten neolabs = ~9% of Anthropic



Source: Anthropic news posts: Series E $61.5B (Mar 2025), Series F $183B (Sep 2025), Series G $380B (Feb 2026), Series H $965B (28 May 2026), all post-money. OpenAI: $157B (Oct 2024) and $300B (Mar 2025) primary rounds (openai.com); $500B employee share sale (Oct 2025, secondary, press; hollow); $852B post-money (openai.com, 31 Mar 2026); talks to raise at least $30B at ~$1.4T as a bridge to a 2027 IPO (Bloomberg, 29 Sep 2026, press; not closed; dashed). Anthropic IPO: investors project about $2T (CNBC, 21 Aug 2026); seeking up to $100B at roughly that valuation, listing expected before the Nov 2026 midterms (Reuters, Sep 2026); confidential draft S-1 (press); not priced, dashed. Neolab cohort = sum of last closed valuations in Exhibit 12 ($90.1B), ours; mixes pre- and post-money as reported. 852 / 157 = 5.4x; 965 / 61.5 = 15.7x; 90.1 / 965 = 9.3%, ours.

SO WHAT

OpenAI 5.4x, Anthropic 15.7x since 2024-25

Anthropic IPO at ~$2T*; OpenAI talks ~$1.4T*

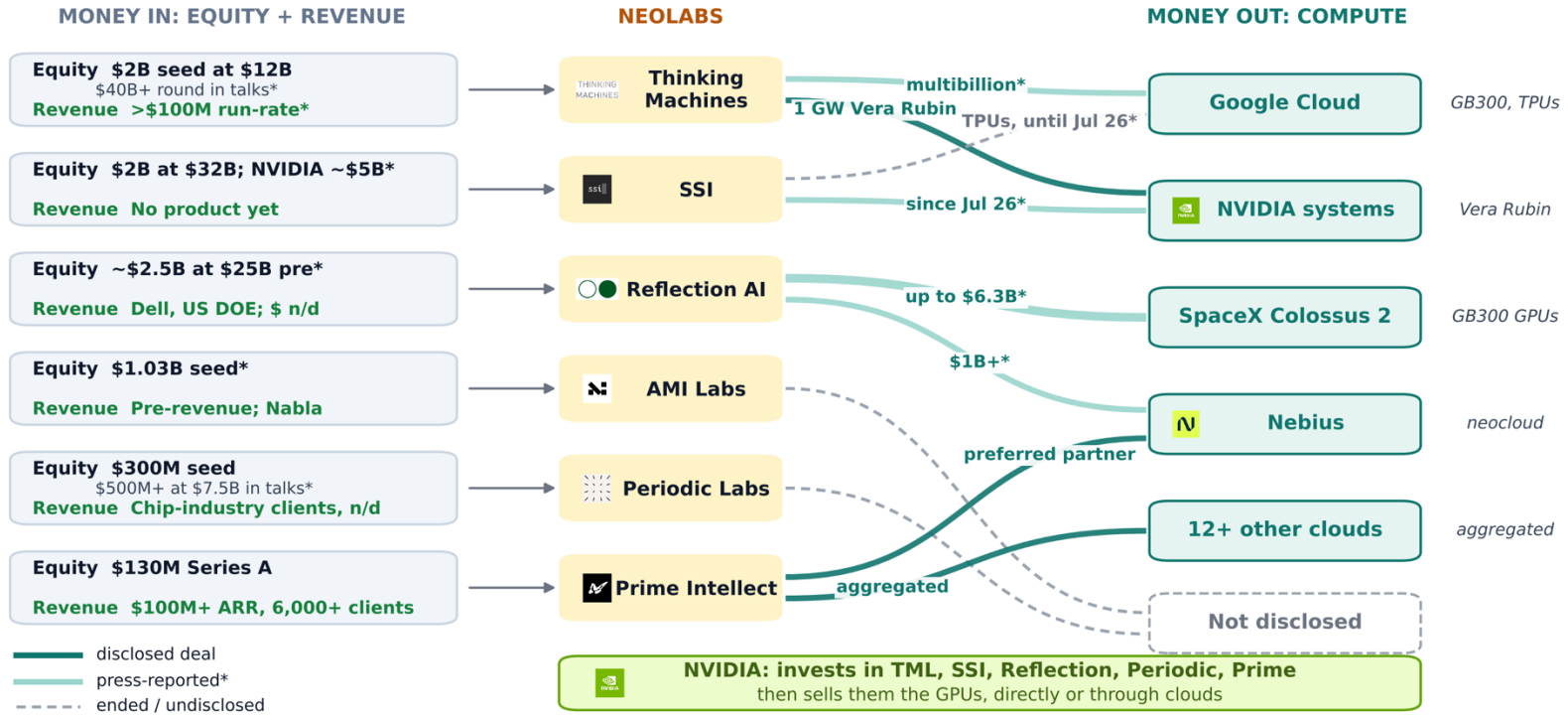
Ten neolabs ≈ 9% of Anthropic

Neolabs run on equity, and spend it on compute

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Exhibit 14B: Follow the money: how neolabs are funded, and where it goes

Investor equity dwarfs revenue at every neolab; most of the money is committed to compute, often with NVIDIA on both sides



Equity: company or press releases (TML a16z-led seed, Jul 2025; SSI Apr 2025 and NVIDIA ~$5B, Jul 2026; Reflection, CEO on CNBC Apr 2026; AMI Mar 2026; Periodic Sep 2025 and Forbes May 2026; Prime Jul 2026). Revenue: TML (TechCrunch); Prime (company, Intel Capital). Compute: Google Cloud (Apr 2026); NVIDIA (Mar 2026); SSI TPU-to-NVIDIA switch (press, Jul 2026); Reflection SpaceX and Nebius (Bloomberg); Prime (Nebius case study; company). Full detail in speaker notes. * = press-reported.

SO WHAT

Equity far exceeds revenue at every lab

Raises become compute deals

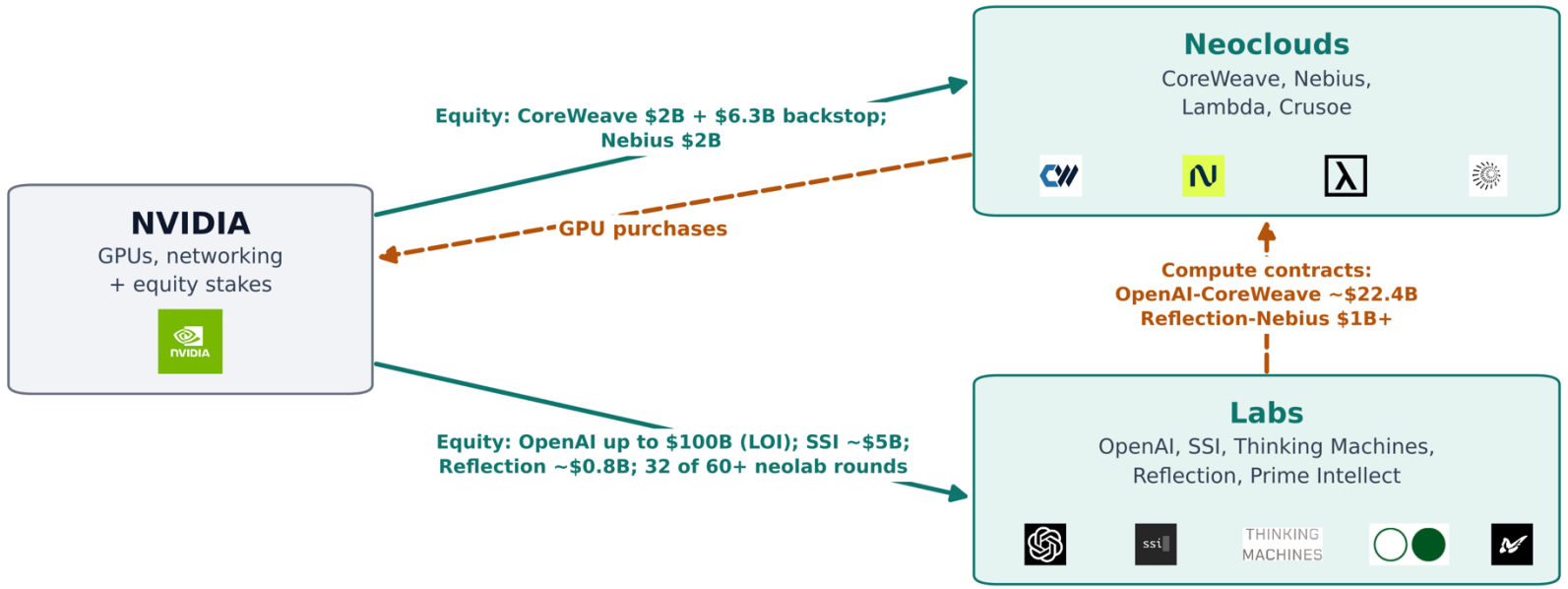
NVIDIA is investor and supplier

NVIDIA funds both sides of its own demand

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Exhibit 14: NVIDIA on both sides: equity out, GPU orders back

NVIDIA invests in the clouds and the labs that end up buying or renting its GPUs: demand partly funded by the supplier



Sources: CoreWeave $2B NVIDIA equity at $87.20/sh (CoreWeave release, 26 Jan 2026) and $6.3B capacity backstop to Apr 2032 (CoreWeave filing via CNBC/DCD); Nebius $2B NVIDIA equity (Q1 2026 letter); OpenAI: NVIDIA 'intends to invest up to $100B' as 10 GW deploys (NVIDIA release, 22 Sep 2025; LOI); SSI ~$5B (Bloomberg, press-reported); Reflection ~$0.8B (press-reported); Thinking Machines undisclosed + 1 GW (NVIDIA blog); NVIDIA in 32 of 60+ neolab rounds (ellipsis Ventures, Aug 2026). Reflection-Nebius $1B+ (TechCrunch); OpenAI-CoreWeave ~$22.4B in three contracts (CoreWeave releases). Schematic; arrows show direction of money.

SO WHAT

Equity in CoreWeave, Nebius, OpenAI, SSI

In 32 of 60+ neolab rounds

Real demand, but partly circular

Why NVIDIA sells through neoclouds

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Exhibit 14C: Why NVIDIA sells through neoclouds, not its own cloud

Its biggest customers make rival chips, owning clusters would wreck its margins, and neoclouds buy only NVIDIA, and buy it first

1 No channel conflict

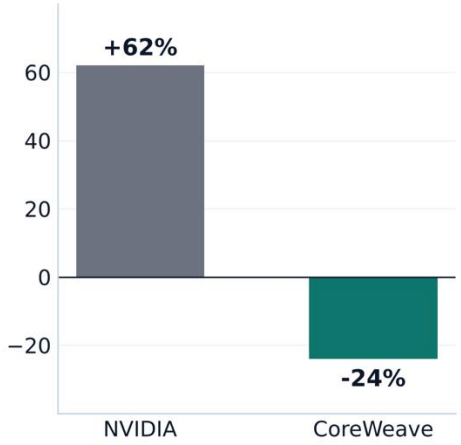
Buyers that make rival AI chips

Google	TPU
Amazon	Trainium
Microsoft	Maia
Meta	MTIA
CoreWeave	none
Nebius	none

NVIDIA scaled back its own DGX Cloud to mainly internal use, stepping back from AWS*

2 Capital stays off its books

Net margin, latest quarter



Of each CoreWeave revenue $: 54¢ depreciation, 25¢ interest. NVIDIA earns 75% gross margin.

3 Loyal, first-in-line buyers

100% NVIDIA
no in-house chip to switch to

First to new chips
CoreWeave: first GB200 (Feb 25) and GB300 (Jul 25)

Europe first
Nebius: first GB300 (Dec 25)

Kept alive by NVIDIA
$2B into CoreWeave, $2B into Nebius, $6.3B capacity backstop

Sources: NVIDIA Q2 FY2027 release (quarter to 26 Jul 2026): revenue $96.2B, GAAP net income $59.7B, gross margin 75.0%; CoreWeave Q2 2026 release: revenue $2,575M, net loss $626M, depreciation $1,393M (54%), interest $640M (25%). Margins = net income / revenue, ours (59.7 / 96.2 = 62%; -626 / 2,575 = -24%). In-house chips: Google TPU, Amazon Trainium, Microsoft Maia, Meta MTIA (company announcements). DGX Cloud scaled back to mainly internal use (The Information via Yahoo Finance, 13 Sep 2025, press). CoreWeave first GB200 NVL72 (Feb 2025) and GB300 NVL72 (Jul 2025), Nebius Europe's first GB300 (Dec 2025) (company claims). NVIDIA equity: CoreWeave $2B (Jan 2026), Nebius $2B (Mar 2026); $6.3B CoreWeave capacity backstop (filing). * = press-reported.

SO WHAT

No conflict: big buyers make rival chips

Capital and debt stay off NVIDIA's books

Neoclouds buy only NVIDIA, and first

AMD buys World Labs after supplying its GPUs

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Exhibit 14D: AMD is buying World Labs, a lab already training on its GPUs

~$8.2B in AMD stock: AMD supplied the chips and the equity first, now it buys the model team to shape its hardware

1 From GPU supplier to owner

Sep 2024
$230M raised at >$1B;
Google Cloud GPUs for training

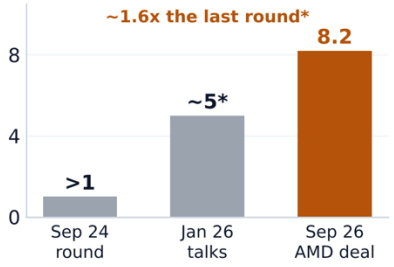
2025
Partnership: training and inference optimised on AMD GPUs

Feb 2026
$1B round: AMD, NVIDIA, Autodesk, Fidelity; ~$5B valuation*

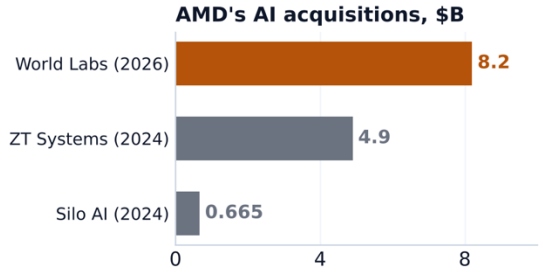
Mar 2026
AMD Ventures follow-on; more compute on AMD Instinct GPUs

28 Sep 2026
AMD to acquire for ~$8.2B in stock; close by end-2026

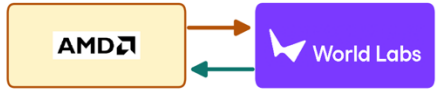
2 World Labs value, $B



AMD's AI acquisitions, $B



3 What each side gets



AMD gives
Instinct GPUs and tuning
Venture equity, then $8.2B stock

AMD gets
World models (Marble, Atlas)
Fei-Fei Li as Chief Scientist
Workload insight for its chips

NVIDIA backed World Labs too, but holds stakes; AMD is buying outright

Sources: AMD press release (28 Sep 2026): ~$8.2B all-stock, close expected end-2026 subject to regulatory approvals, Fei-Fei Li to be EVP and Chief Scientist reporting to Lisa Su; World Labs blog (28 Sep 2026): partnership began 2025 with 'model training and inference optimization on AMD GPUs'; AMD Ventures page (19 Mar 2026): follow-on investment after Series B, compute 'on AMD Instinct GPUs'. Funding: $230M at >$1B (TechCrunch, Sep 2024); $1B raise with AMD, NVIDIA, Autodesk, Fidelity (World Labs, 18 Feb 2026); ~$5B valuation in Jan 2026 talks (Bloomberg*). AMD deals: Silo AI $665M (AMD, Jul 2024), ZT Systems $4.9B (AMD, Aug 2024). Multiple = 8.2 / 5, ours. * = press-reported.



Signing the AMD–World Labs deal

AMD sold the chips, then bought the lab

First neolab exit: to a chipmaker

SECTION 4 OF 4

Risks & challenges

What could break, explained technically

COMING UP

Power, cooling and the grid 46–47

GPU obsolescence and failures 48–49

Leverage and concentration 50

Chip risk: Cerebras, TPUs, Trainium 51–52

Commoditisation: China's open models, copycats 53–55

China's labs: IPOs, prices, state support 56–57

NVIDIA's own open models: Nemotron 58

Where neolabs can still win 59–61

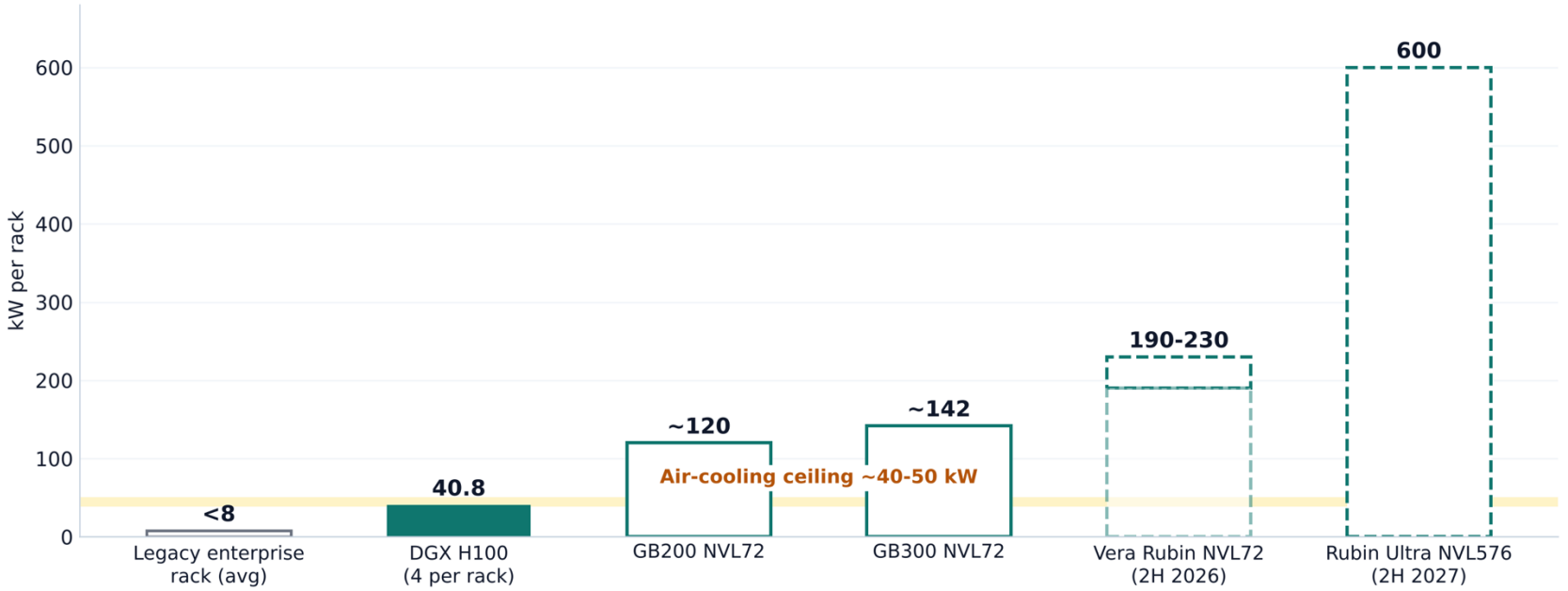
ARC view and sources 62–63

A GB200 rack draws 15x a legacy rack

ARC RESEARCH | AI UPDATE | RISKS: POWER & COOLING

Exhibit 15: Power per rack by system generation (kW)

Racks went from <8 kW to 120+ kW and are heading to 600 kW: that forces liquid cooling and new buildings



Source: legacy average <8 kW, Uptime Institute 14th annual survey (2024); DGX H100 4 systems x 10.2 kW = 40.8 kW, NVIDIA DGX SuperPOD H100 design guide; GB200 NVL72 ~120 kW and GB300 NVL72 up to ~142 kW, NVIDIA reference figures as relayed by secondary sources (pantheon.run; HPE quotes 132 kW nominal); Vera Rubin NVL72 190-230 kW Max-Q/Max-P (HashrateIndex; another source says 120-130 kW, unsettled); Rubin Ultra NVL576 600 kW, Jensen Huang, GTC 2025 keynote via DCD (2H 2027). Air-cooling ceiling ~40-50 kW is approximate (vendor and Uptime sources). Hollow = secondary source; dashed = future system.

SO WHAT

Air cooling tops out near 40-50 kW

New racks need liquid to the chip

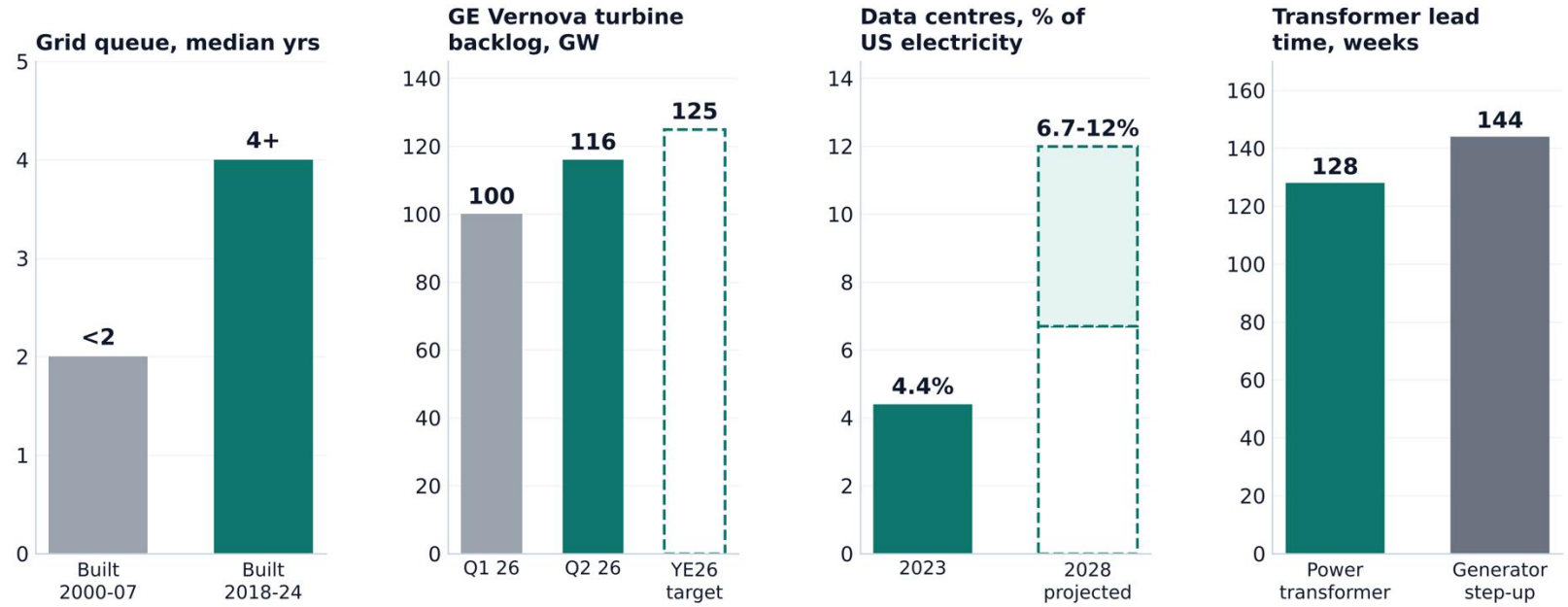
Most existing halls can't host them

Power, not chips, is the binding constraint

ARC RESEARCH | AI UPDATE | RISKS: POWER & GRID

Exhibit 16: Four measures of the power bottleneck

Grid connections take twice as long, turbines are booked to 2031, and transformers take ~2.5 years



Source: (1) LBNL 'Queued Up' 2025 edition: median time from interconnection request to commercial operation, projects built 2000-07 (<2 yrs) vs 2018-24 (>4 yrs); all generation/storage, not only data centres; bars drawn at the stated bounds. (2) GE Vernova gas-turbine backlog incl. slot reservations, Q1 and Q2 2026 calls via Utility Dive; 125 GW = year-end 2026 target. (3) LBNL Dec 2024 US data-centre energy report: 4.4% of US electricity in 2023, 6.7-12% projected for 2028. (4) Wood Mackenzie Q2 2025 survey via POWER magazine: lead times for power and generator step-up transformers. Dashed = projection or target.

SO WHAT

Grid hook-up: 4+ years, median

Gas turbines booked out to 2031

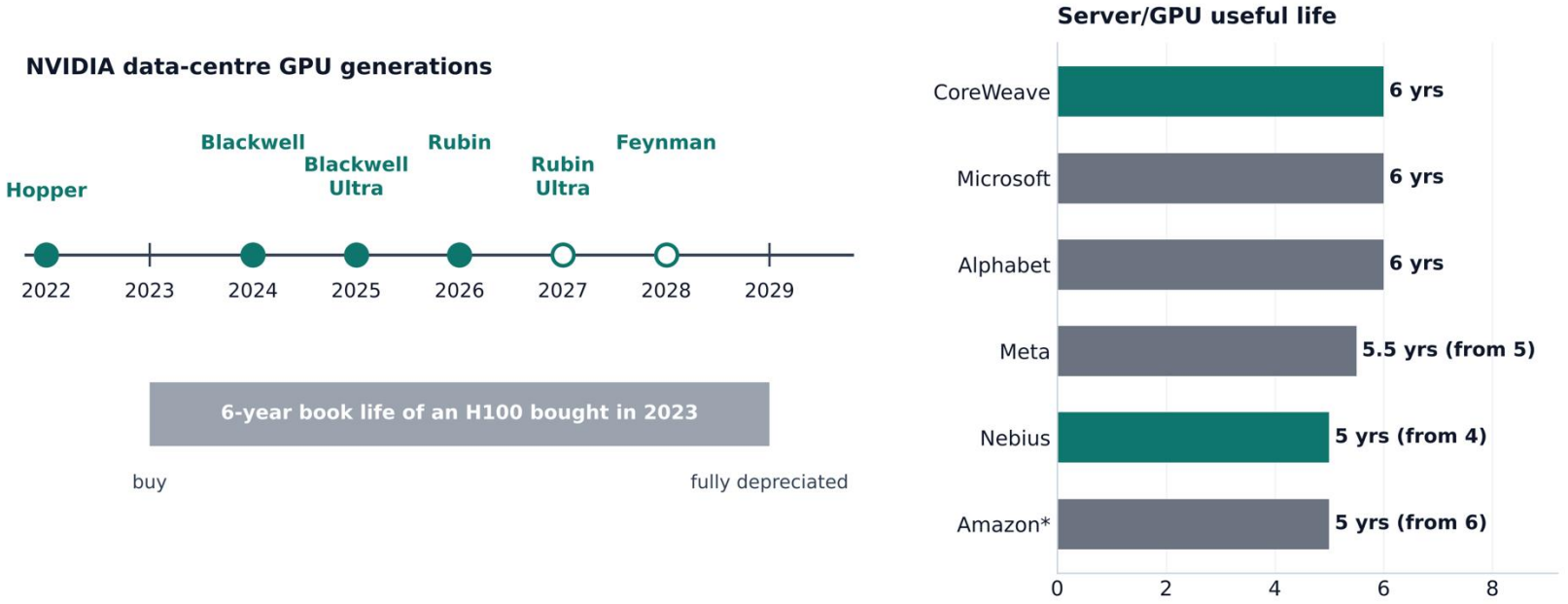
Whoever has power has customers

Chips turn over yearly; books assume 6 years

ARC RESEARCH | AI UPDATE | RISKS: GPU OBSOLESCENCE

Exhibit 17: NVIDIA cadence vs the useful life on the books

A GPU bought in 2023 is depreciated to 2029, by which time four newer generations will have shipped



Source: NVIDIA launch announcements (Hopper GTC 2022, Blackwell GTC 2024); Blackwell Ultra 2H 2025, Rubin full production Jun 2026, Rubin Ultra 2H 2027, Feynman 2028 per NVIDIA GTC roadmaps as relayed by secondary sources. Useful lives: Amazon 10-Q Q1 2025 (6 -> 5 yrs for a subset of servers, 2025); Meta FY2025 10-K (5.5 yrs); Microsoft and Alphabet 6 yrs (FY2023 changes, secondary); CoreWeave 6 yrs (S-1, secondary); Nebius 4 -> 5 yrs from Q1 2026 (Q1 2026 shareholder letter). 6-yr life of a 2023 purchase = illustrative, ours. Resale: used H100 ~45% of new in year 3 (Forbes, Jun 2026, press-reported).

SO WHAT

New NVIDIA generation about every year

Used H100 ~45% of new by year 3

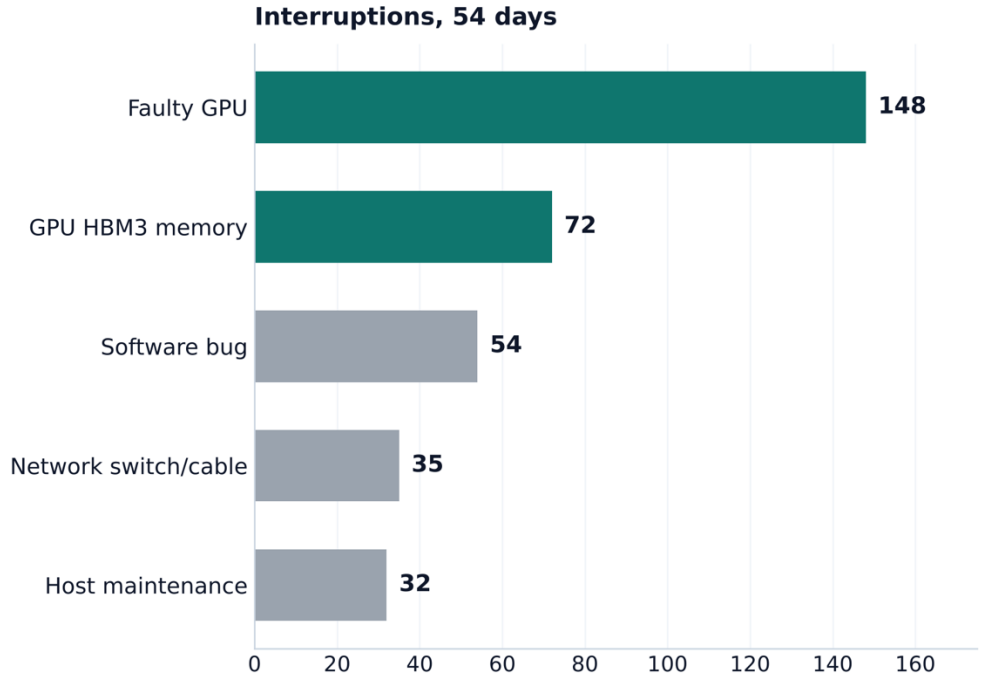
Shorter lives would cut reported profit

At 16k GPUs, something breaks every 3 hours

ARC RESEARCH | AI UPDATE | RISKS: CLUSTER RELIABILITY

Exhibit 18: Meta Llama 3 training run: unexpected interruptions by root cause

On 16,384 H100s, 419 unexpected failures in 54 days: one every ~3 hours, mostly GPUs and their memory



~3.1 hours

between unexpected failures

>90%

effective training time

38-43%

model FLOPs utilisation (MFU)

Source: Meta, 'The Llama 3 Herd of Models', arXiv 2407.21783 (Jul 2024), Table 5 and Section 3.3.4: 466 interruptions in a 54-day snapshot (47 planned, 419 unexpected), ~78% of unexpected attributed to confirmed hardware issues, >90% effective training time, MFU 38-43% (BF16). Top five categories shown, counts as reported. Hours per failure = 54 x 24 / 419 = 3.1, ours.

SO WHAT

GPUs and HBM: 220 of 419 failures

Fast restarts keep >90% productive

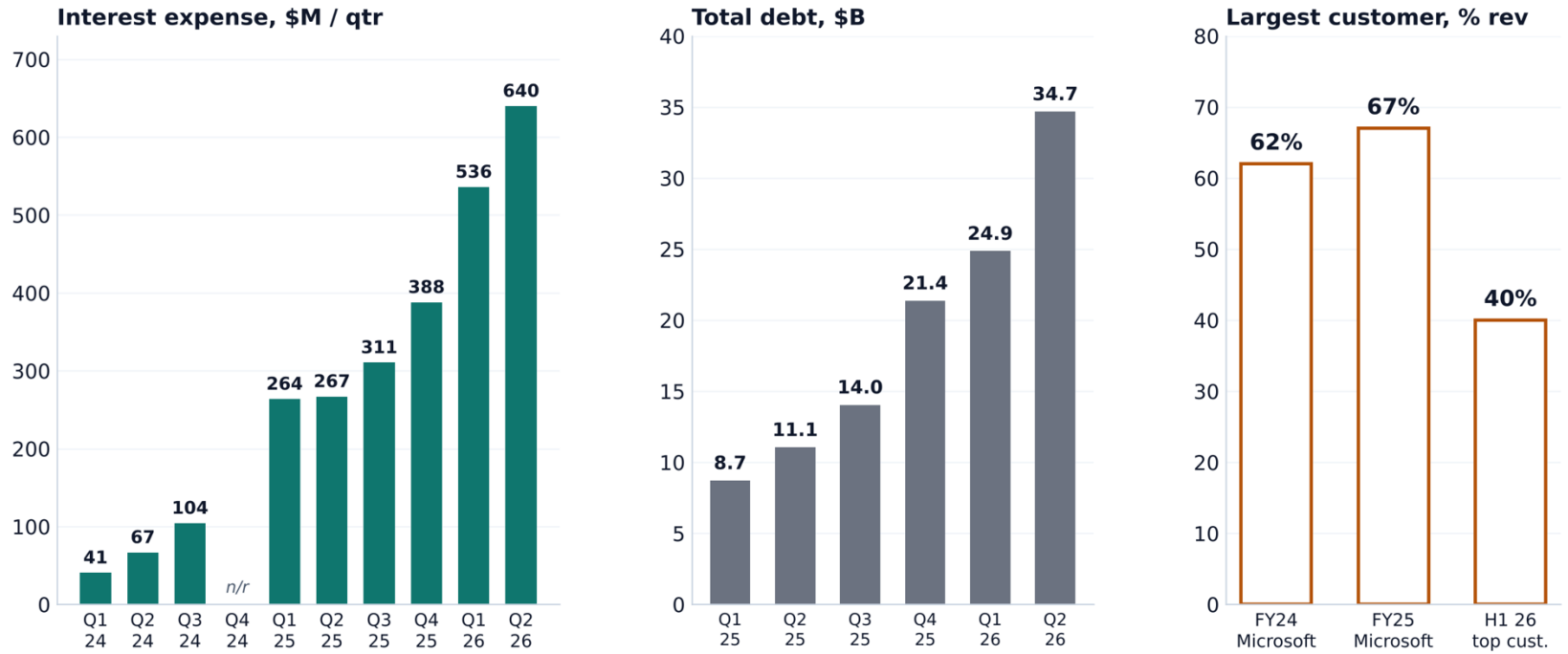
Only ~40% of peak compute is used

Leverage and concentration are the pressure points

ARC RESEARCH | AI UPDATE | RISKS: FINANCING

Exhibit 19: CoreWeave: interest cost, debt and customer concentration

Quarterly interest is up ~10x in two years; debt quadrupled in 15 months; one customer is still ~40%



Source: CoreWeave quarterly earnings releases (interest expense; Q4 2024 not reported separately, shown as gap) and balance sheets (total debt; Q2 2026 $34.7B per release, $35.6B principal per 10-Q). Concentration: FY2024 Microsoft 62% (S-1/10-K via TNW, press-reported); FY2025 ~67% (10-K, via secondary summary); H1 2026 largest customer 40%, unnamed (10-Q, via secondary summary). Hollow = press/secondary. 640 / 66.8 = 9.6x (Q2 26 vs Q2 24); 34.7 / 8.7 = 4.0x (Q2 26 vs Q1 25), ours.

SO WHAT

Interest up 9.6x in two years

EBITDA covers interest only 2.4x

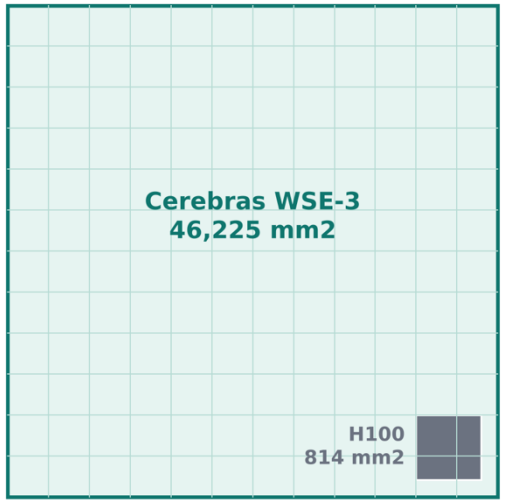
Top customer still ~40% of revenue

An all-NVIDIA fleet is a single-supplier bet

ARC RESEARCH | AI UPDATE | RISKS: NVIDIA DEPENDENCE

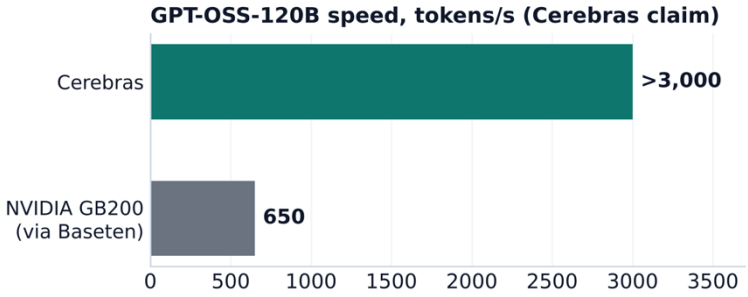
Exhibit 6D: The alternatives to NVIDIA: Cerebras's wafer and AMD's GPUs

Cerebras keeps model weights on one giant chip; AMD packs more memory per GPU than NVIDIA. OpenAI and Anthropic now buy both



Drawn to scale (area)

	Cerebras WSE-3	AMD MI355X	AMD MI455X (2H 2026)	NVIDIA H100
Memory	44 GB SRAM	288 GB HBM3E	432 GB HBM4	80 GB HBM3
Bandwidth	21 PB/s	8 TB/s	23.3 TB/s	3.35 TB/s
Who buys	OpenAI 750 MW	Anthropic, Crusoe	OpenAI 6 GW, Anthropic 2 GW*	everyone



Sources: Cerebras WSE-3 datasheet (company): 46,225 mm2, 4T transistors, 44 GB on-chip SRAM, 21 PB/s; NVIDIA H100 (company): 814 mm2 die, 80 GB HBM3, 3.35 TB/s; AMD MI355X product page (company): 288 GB HBM3E, 8 TB/s, 185B transistors, 1,400 W; AMD MI400 page (company): MI455X 432 GB HBM4, 23.3 TB/s, Helios racks of 72 GPUs expected 2H 2026. Deals: OpenAI-AMD up to 6 GW with warrants (company); AMD-Anthropic up to 2 GW of MI450 from 1H 2027 plus up to $5B AMD equity investment (Tom's Hardware, 22 Jul 2026, press); OpenAI-Cerebras 750 MW (company). Speed: Cerebras blog (6 Nov 2025), Cerebras-selected comparison. Area to scale; 46,225 / 814 = 57x, ours.

SO WHAT

Neocloud fleets are almost all NVIDIA

AMD: more memory per GPU; OpenAI, Anthropic buy

One vendor sets supply, price, roadmap

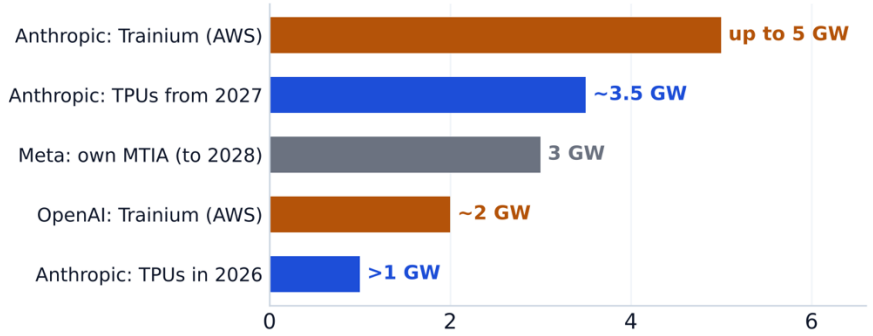
TPUs and Trainium pull demand from NVIDIA

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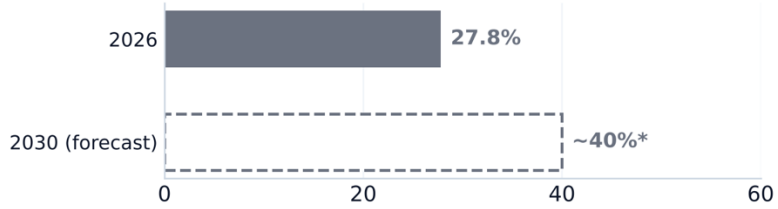
Exhibit 20I: Google and Amazon chips are pulling demand away from NVIDIA

The biggest labs now commit gigawatts to TPUs and Trainium; neoclouds sell only NVIDIA, and neolabs queue behind the clouds' own labs

1 Custom-chip commitments by big labs, GW



2 Custom chips' share of AI server shipments



3 Why it is a risk

Neoclouds: one supplier

100% NVIDIA, while Google puts TPUs in rival hosts (Fluidstack, a Blackstone JV)

Neolabs: back of the queue

Clouds serve their own labs first; AWS 2027 capacity 'largely reserved'; Google 'supply constrained'

Full-stack rivals

Google and Amazon own the chip, the cloud and the software (XLA/JAX, Neuron)

For now: still NVIDIA

Neocloud demand '99 percent' NVIDIA; no neolab is a disclosed TPU/Trainium customer

Sources: Broadcom 8-K (6 Apr 2026: Anthropic access to ~3.5 GW of Google TPUs from 2027); Anthropic (23 Oct 2025: up to 1M TPUs, well over 1 GW in 2026); Amazon (20 Apr 2026: Anthropic up to 5 GW Trainium; 27 Feb 2026: OpenAI ~2 GW Trainium); Meta/Broadcom MTIA 3 GW through 2028 (Broadcom call, 3 Jun 2026); Amazon Q2 2026 call (chips run-rate >$25B incl. Graviton and Nitro; 2027 capacity 'largely reserved'); Alphabet Q2 2026 call (TPU systems delivered to customer data centres; 'supply constrained'); Google-Blackstone TPU cloud (Google, 19 May 2026); TrendForce (Jan and Mar 2026: ASIC share of AI server shipments); DCD (5 May 2026: Nebius '99 percent' NVIDIA demand). * = press-reported or forecast.

SO WHAT

Big labs commit GWs to TPU, Trainium

Neoclouds sell only NVIDIA

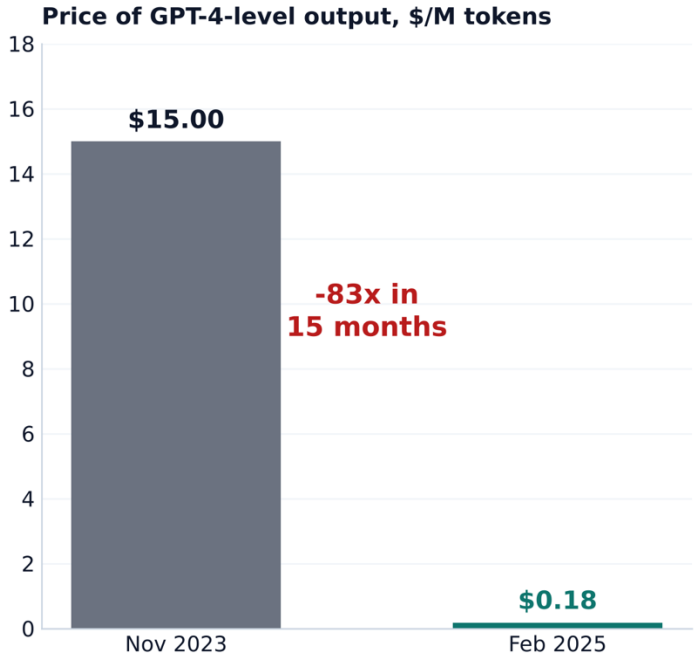
Neolabs wait behind clouds' own labs

The model layer commoditises fast

ARC RESEARCH | AI UPDATE | RISKS: NEOLABS

Exhibit 20: Why the model layer commoditises fast

Capability gets cheap quickly and open models catch up in months, while each frontier run costs more



~4 months

open-weight lag to closed frontier

~40x / yr

price drop at fixed GPQA score

2.4x / yr

frontier training-run cost growth

3 leads

left Thinking Machines for OpenAI, Jan 2026

Source: Epoch AI 'LLM inference prices' (Mar 2025): cheapest price for GPT-4-level MMLU performance, $15.00 (Nov 2023) vs $0.18 (Feb 2025) per M tokens; price to reach a fixed GPQA score fell ~40x/yr. Epoch AI 'open-closed ECI gap' (May 2026): open-weight models ~4 months behind the closed frontier since Jan 2026. Cottier et al., arXiv 2405.21015 (2024): frontier training cost grew 2.4x/yr since 2016. Talent: Fortune (16 Jan 2026), three Thinking Machines co-founders/leads returned to OpenAI. 15 / 0.18 = 83x, ours.

SO WHAT

GPT-4-level output: 83x cheaper

China's open models: free, ~4 months behind

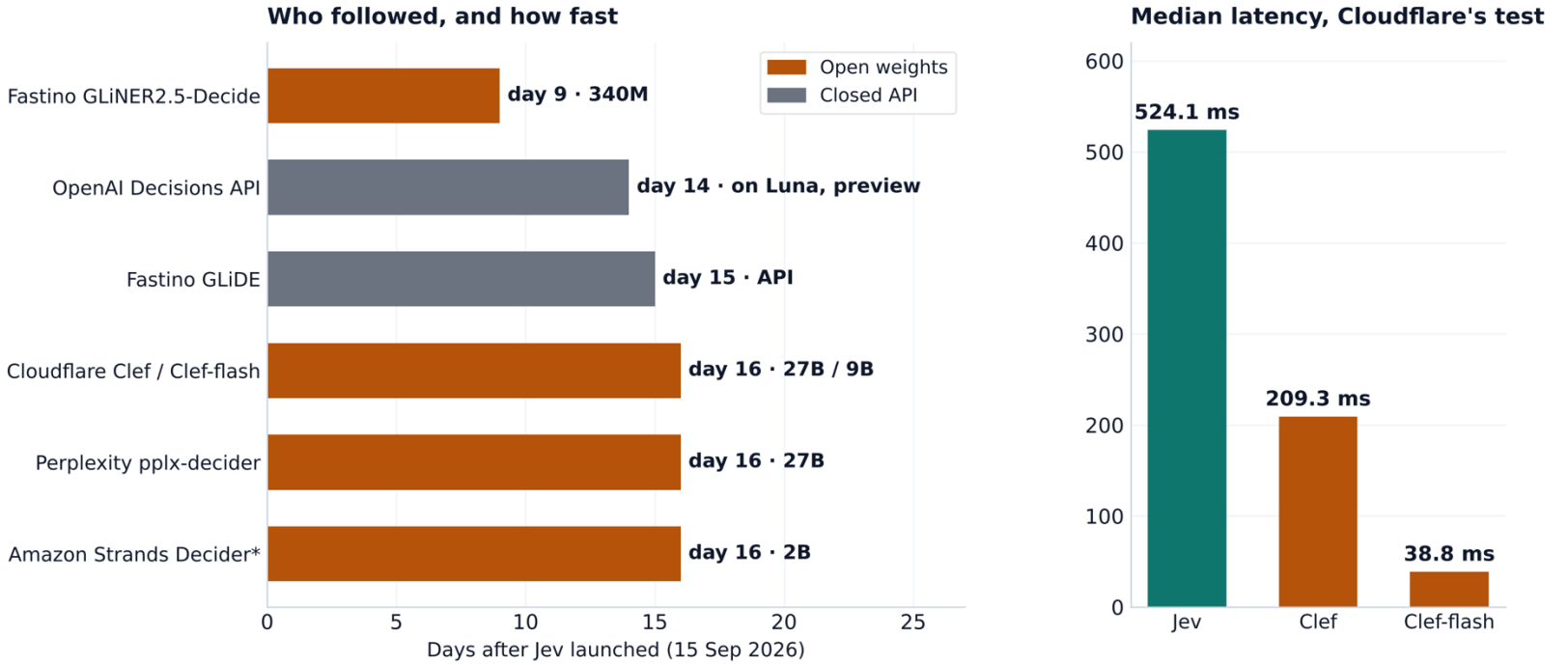
Each frontier run costs ~2.4x more

A new idea gets copied in 16 days

ARC RESEARCH | AI UPDATE | RISKS: NEOLABS

Exhibit 20B: Case study: TypeSafe's Jev was matched within 16 days

OpenAI, Cloudflare, Perplexity, Amazon and Fastino all shipped decision models within 16 days of Jev's 15 Sep launch, four of them open-weight



Sources: TypeSafe blog (15 Sep 2026, company): Jev, a 'System One' decision model, closed API. OpenAI DevDay 2026 recap (29 Sep, company): Decisions API on Luna, limited preview. Cloudflare blog (1 Oct, company): Clef (Qwen 27B base) and Clef-flash (Qwen 9B), Apache 2.0, on Workers AI and Hugging Face; median latency in Cloudflare's own test: Clef-flash 38.8 ms, Clef 209.3 ms, Jev 524.1 ms. Perplexity pplx-decider-v1-27b model card (company): Apache 2.0, 85.71% vs Jev 84.51% on Perplexity's panel; launch date 1 Oct per press. Fastino GLiNER2.5-Decide (24 Sep, 340M, open) and GLiDE (30 Sep, closed): MarkTechPost (press). Amazon Strands Decider 2B (1 Oct, open): Digital Applied (press). Days after 15 Sep, ours.

SO WHAT

OpenAI, Cloudflare, Perplexity followed in ≤16 days

Cloudflare claims 2.5x faster, open weights

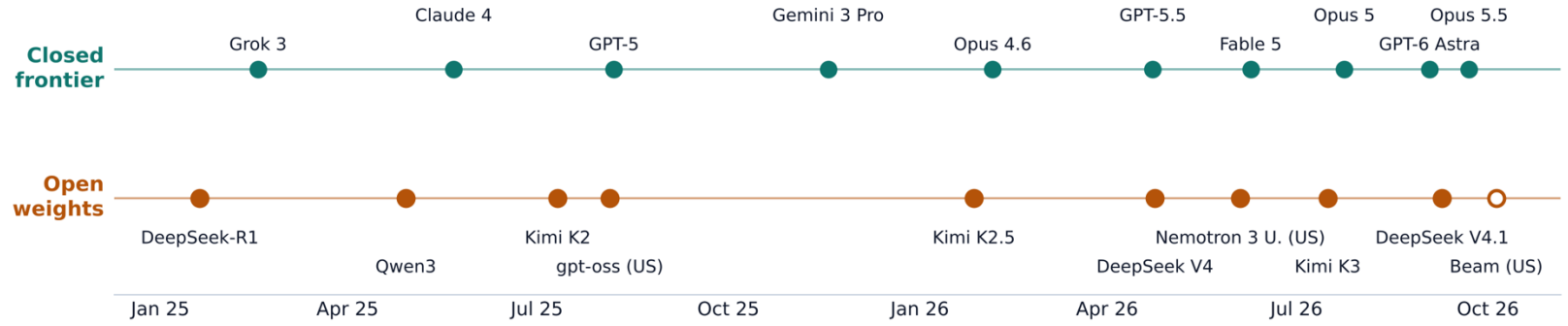
Lead time is weeks, not years

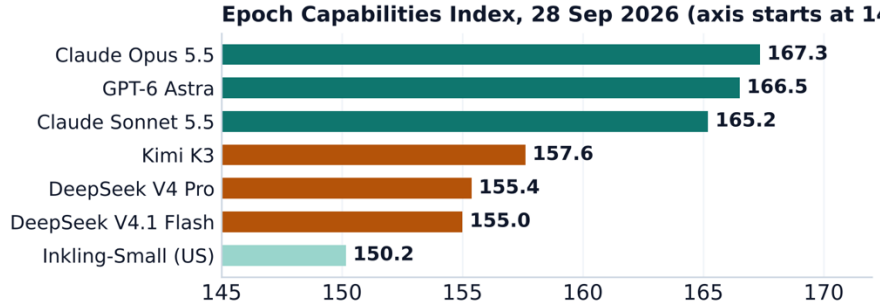
Open models trail the frontier by ~4 months

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Exhibit 20C: Frontier releases vs open-weight releases, Jan 2025 to Oct 2026

Closed labs ship every 1-3 months; Chinese open models follow about 4 months behind, and the best US open model trails both





~4 months
open lag to the frontier (Epoch)

~10 pts
top-3 closed vs top-3 open

-7.5 pts
best US open vs best China open

Release dates: company pages and press; Beam (Reflection, 5 Oct 2026) hollow: weights due later in Oct. Capability: Epoch AI Capabilities Index (eci_scores.csv, latest entry 28 Sep 2026, primary); Epoch classes Qwen 3.8 Max and GLM-5.3 as closed (API-only when scored). Lag: Epoch 'open-closed ECI gap' (29 May 2026, its latest published estimate). Gaps, ours: top-3 closed 166.35 - top-3 open 156.00 = 10.4; Kimi K3 157.61 - Inkling-Small 150.16 = 7.5. Releases shown are a selection.

SO WHAT

A frontier release every 1-3 months

China's open models ~10 pts behind

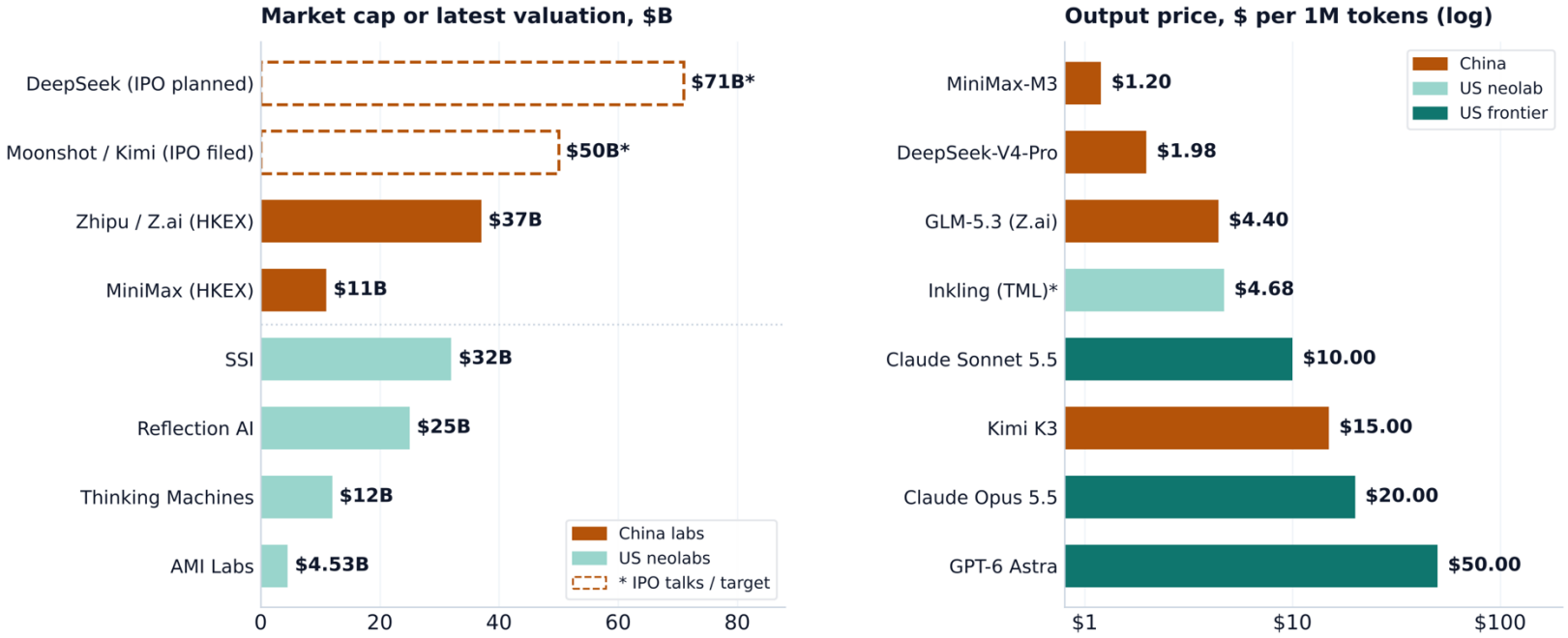
Best US open model trails China's

China's open labs go public, and price lower

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Exhibit 20D: China's open-model labs go public and sell tokens cheaper

Zhipu and MiniMax listed in Jan 2026; DeepSeek and Moonshot are preparing IPOs. Public money funds a price war



Valuation: Zhipu HKEX 2513 (listed 8 Jan 2026) HK$291.7bn ~US$37B and MiniMax HKEX 0100 (listed 9 Jan 2026) HK$85.3bn ~US$11B at 2 Oct 2026 close (stockanalysis.com, Simply Wall St; aggregators); DeepSeek ~US$71B in talks, IPO filing targeted late 2026-27 (Bloomberg via TechCrunch, press); Moonshot confidential HK filing targeting ~US$50B (LatePost, press); US neolabs as on the valuations slide. Zhipu follow-ons ~US$4B (Jul) + ~US$5B (Sep) (press). Prices: official pricing pages, 6 Oct 2026, output $ per 1M tokens, standard tier: MiniMax-M3 1.20 (labelled 50% off); DeepSeek-V4-Pro 1.98 off-peak (3.96 peak); GLM-5.3 4.40; Kimi K3 15; Claude Sonnet 5.5 10, Opus 5.5 20; GPT-6 Astra 50; Inkling on Tinker 4.68 (third-party listing, press). 20 / 1.98 = 10x; 50 / 1.20 = 42x, ours. Dashed = talks/target.

SO WHAT

Zhipu, MiniMax listed in Hong Kong

DeepSeek, Moonshot preparing IPOs*

Their tokens cost 2-40x less

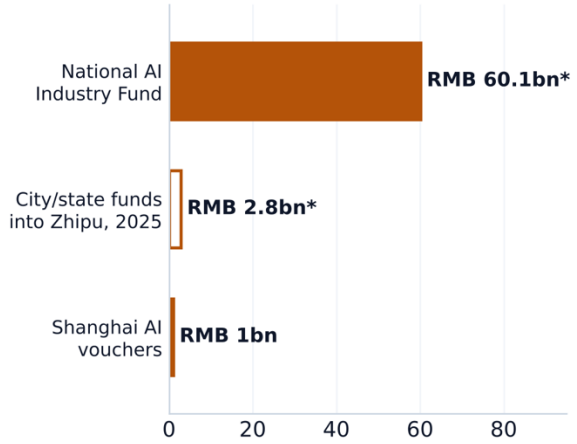
State money, talent and engineering keep costs low

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Exhibit 20E: Why China's labs can price low: state, talent, engineering

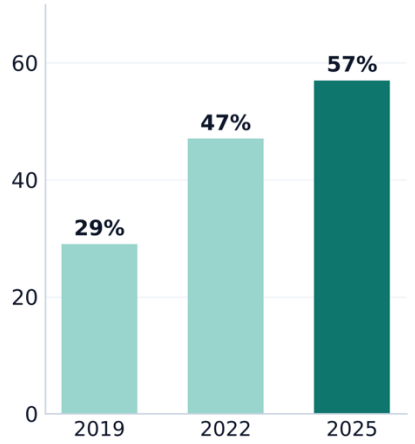
Government funds and compute vouchers lower costs, China now trains most top AI researchers, and its labs publish efficiency tricks

1 State money (RMB bn)



Shanghai vouchers cover up to 30% of compute rent; 'AI+' plan targets 70% agent adoption by 2027

2 Top AI researchers with a Chinese undergrad degree



41% of top researchers now work in China, vs 34% in the US (2025)

3 Efficiency engineering

DeepSeek-V3
2.79M H800 GPU-hours, ~$5.6M final run; FP8, MLA, MoE

Open infra
FlashMLA, DeepEP, 3FS released as open source

MiniMax-01
lightning attention, 1M-token training context

Kimi K2
MuonClip optimizer; 32B of 1T params active

GLM-5
adapted to 7 Chinese chip platforms incl. Huawei Ascend

Funding: National AI Industry Investment Fund ~RMB 60.1bn (Jan 2025; Global Trade Alert/SCMP, press); state and city funds into Zhipu in 2025: Hangzhou 1.0bn, Zhuhai Huafa 0.5bn, Chengdu Hi-Tech 0.3bn (Mar, Reuters), Shanghai Pudong VC + Zhangjiang 1.0bn (Jul, DealStreetAsia) (press); Shanghai vouchers RMB 0.6bn compute (up to 30% of rent) + 0.3bn model + 0.1bn data (Shanghai govt, Jul 2025, primary); State Council 'AI+' plan (Aug 2025, primary). Talent: Chinese-undergrad share of top AI researchers 29% (2019), 47% (2022) (MacroPolo Global AI Talent Tracker 2.0) and 57% (2025) (Carnegie, Sep 2026); 41% work in China vs 34% in the US (Carnegie). Engineering: DeepSeek-V3 report (arXiv 2412.19437); MiniMax-01 (2501.08313); Kimi K2 (2507.20534); GLM-5 (2602.15763).

SO WHAT

State funds and compute vouchers

57% of top AI researchers studied in China

Efficiency tricks, published openly

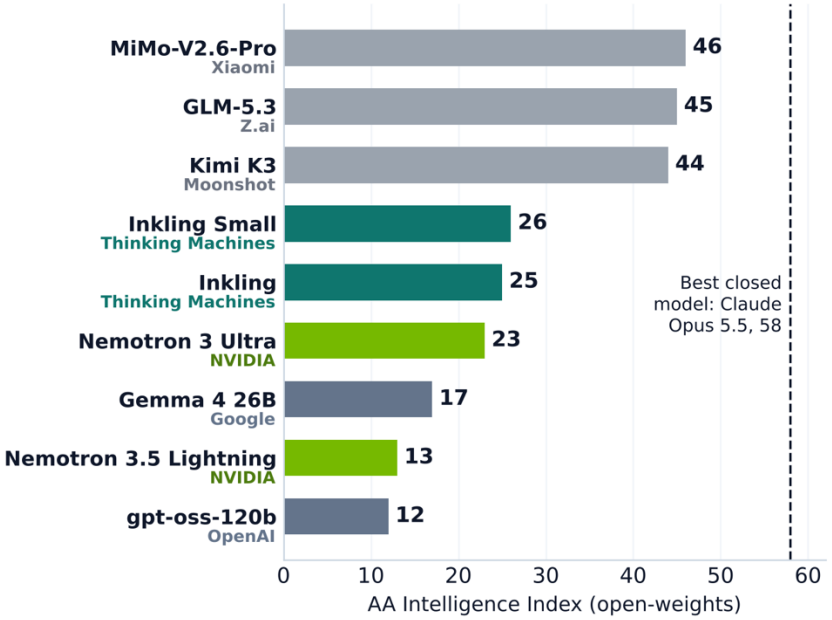
NVIDIA now gives away its own open models

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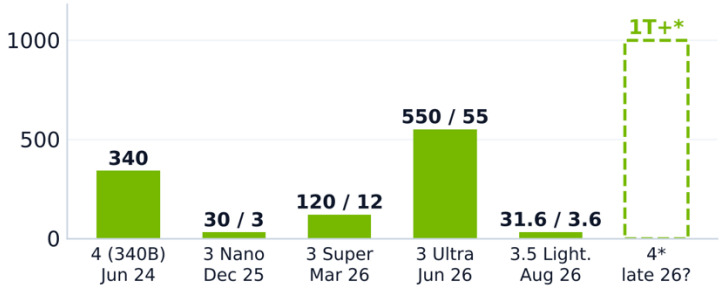
Exhibit 20F: NVIDIA now gives away its own open models: Nemotron

Supplier, investor and now rival: Nemotron 3 Ultra is free and open, though it still trails China's best and NVIDIA-backed Inkling

1 How good: Nemotron vs other open models



2 Bigger releases (params, B: total / active)



3 Why it hurts neolabs

Free and fully open
weights, >10T tokens of data, recipes, open licence

Deep pockets
~$26B over 5 yrs*; $6B Poolside licence + ~100 staff*

Built-in reach
NIM, 25+ platforms, CoreWeave, Nebius; used by Perplexity

Sources: Artificial Analysis Intelligence Index leaderboard, open-weights models, fetched 6 Oct 2026 (index v4.3; scores not comparable with pre-Sep 2026 versions); Claude Opus 5.5 = top closed model on the same page. Nemotron specs: NVIDIA newsroom (Nemotron 3, 15 Dec 2025), NVIDIA blog (Super, 11 Mar 2026; Lightning, 11 Aug 2026), Hugging Face model card (Ultra 550B-A55B, 4 Jun 2026); Nemotron-4 340B (Jun 2024). Nemotron 4 1T+ (The Information, Aug 2026*). $26B over five years on open models (Wired, Mar 2026*, linked to multi-year cloud commitments in NVIDIA's 10-Q); $6B Poolside licence and ~100 engineer offers (WSJ, Aug 2026*). >10T tokens of open data (NVIDIA, Super). * = press-reported or unconfirmed.

SO WHAT

Supplier, investor and now rival

Free US open model caps neolab pricing

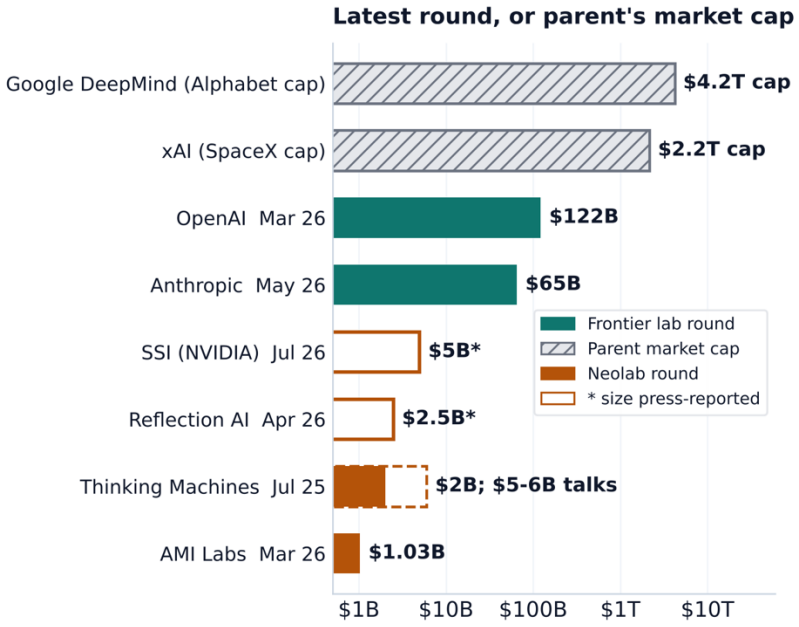
Still trails China and Inkling, for now

Neolabs can't subsidise, so they must specialise

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Exhibit 20A: Neolabs can't outspend the big labs, so they must pick a lane

Frontier labs raise 13-118x a neolab's round, and DeepMind and xAI sit inside $2-4T parents: neolabs cannot subsidise free access



Six lanes where a neolab can charge; two stand out

US open-weight alternative
trusted open models vs China's DeepSeek, Qwen, Kimi, GLM
Reflection; TML's Inkling

Domain expert models
one model per domain, trained on that field's own data
Periodic (materials); AMI + Nabla

Sovereign / on-prem
models customers run on their own hardware
Reflection: Dell, DOE, Korea

Customisation tools
fine-tuning and RL as a paid service
TML Tinker, Prime Intellect

Proprietary data
own experiments, not scraped web
Periodic robotic labs

New paradigms
beyond next-token LLMs
AMI world models

Most recent primary round as of 5 Oct 2026. Sources: OpenAI $122B committed at $852B post (openai.com, 31 Mar 2026); Anthropic Series H $65B (anthropic.com, 28 May 2026); SSI: NVIDIA $5B equity (Jul 2026; amount press-reported); Reflection AI ~$2.5B at $25B pre-money (CEO on CNBC, Apr 2026; size press-reported); Thinking Machines $2.0B seed (Jul 2025; $5-6B round in talks per The Information, Sep 2026, not signed); AMI Labs $1.03B (Mar 2026). Hollow = amount press-reported only; dashed = in talks. 65/5 = 13x, 122/1.03 = 118x, ours. Market caps at 2 Oct 2026 close (stockanalysis.com, aggregator): Alphabet ~$4.2T; SpaceX ~$2.16T (Nasdaq: SPCX; includes xAI, merged Feb 2026). A market cap is not a funding round; shown to indicate the parent's resources. The six lanes are an ARC Research view, not company guidance; the two highlighted lanes are the ones we think best fit a neolab facing free Chinese open-weight models.

SO WHAT

China's free open models set the price

Big labs raise 13-118x more per round

Best bets: US open weights, domain experts

Data wall: domain experts + RL are the moat

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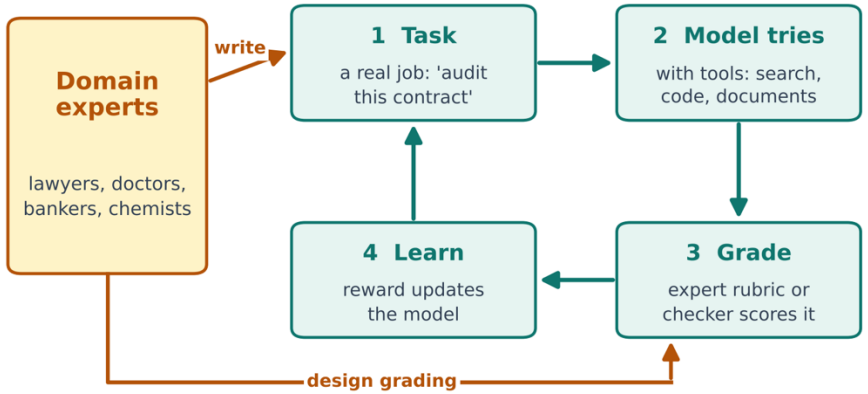
Exhibit 20G: Data wall: domain experts + RL become the new moat

Public text is running out, so gains shift to RL: experts write the tasks and grading, and a small open model wins its field

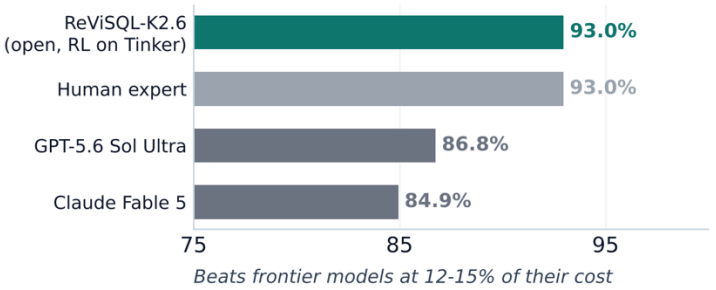
1 Why now: pretraining is hitting a data wall



2 How an RL environment works

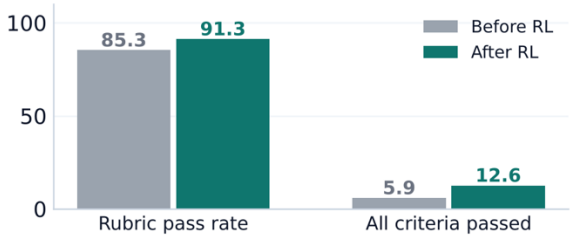


3 Code: text-to-SQL accuracy



Beats frontier models at 12-15% of their cost

4 Law: Harvey agent, % (GLM-5.1)



Sources: Epoch AI, Will we run out of data? (2024: ~300T tokens of public human text, 90% CI 100T-1,000T, fully used 2026-2032); RL compute 10x per step: OpenAI o1 to o3 and xAI Grok 3 to 4 (Epoch AI, May 2025). Thinking Machines with UIUC and Bridgewater AIA Labs, 'Putting Task Expertise into RL' (27 Aug 2026): ReViSQL-K2.6 trained on Tinker, BIRD text-to-SQL, 16-sample result, 'at 12-15% of their cost'; Harvey and Applied Compute (22 Jun 2026): GLM-5.1 post-trained on Harvey's Legal Agent Benchmark, rubric pass rate and all-pass rate; All results are company-reported, not independently verified.

SO WHAT

Public text runs out 2026-32*

Small open model beats frontier

Moat: environments, not weights

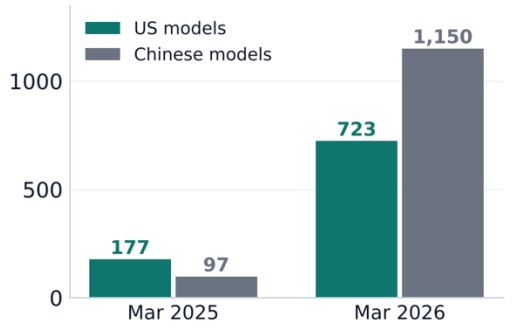
A trusted US open model is an opening

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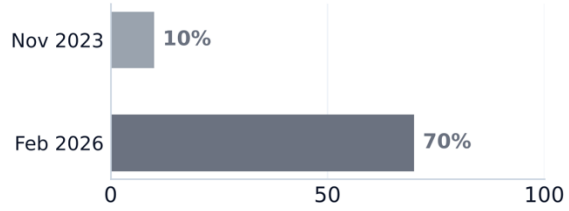
Exhibit 20H: The US open-model gap is an opening for neolabs

China now leads open-model downloads; Washington and NVIDIA want a trusted US alternative, and neolabs can be it

1 Open-model downloads, M



New fine-tunes built on Chinese models



2 Who is pushing for a US alternative

US government

AI Action Plan
'Encourage open-source and open-weight AI' (Jul 2025)

DOE Genesis
open science models (Aug 2026); Reflection serves 17 labs*

Funding
NSF + NVIDIA $152M for Ai2 open models

Curbs on China
DeepSeek banned at Commerce, Navy, NASA*

NVIDIA

Nemotron
its own US open models

Coalition
Reflection, Thinking Machines, Mistral, Perplexity and others

Money
~$26B on open models*

Compute
1 GW for Thinking Machines; stakes in Reflection, TML

3 Opportunity for neolabs: be the trusted US open model

for governments, regulated firms and allies that won't run Chinese weights (Reflection, Thinking Machines)

Sources: The ATOM Report (Lambert and Brand, arXiv 2604.07190, Apr 2026): cumulative Hugging Face downloads of open models by country; share of new fine-tuned models built on Chinese bases. America's AI Action Plan (White House, Jul 2025). DOE Genesis Open Models Initiative (energy.gov, 7 Aug 2026). Reflection as model provider to DOE's 17 national labs (Axios, 22 May 2026*). NSF and NVIDIA $152M for Ai2's open models (Ai2, Aug 2025). DeepSeek bans at Commerce, Navy and NASA (press*). NVIDIA Nemotron Coalition (NVIDIA, 16 Mar 2026); ~$26B on open models (Wired, Mar 2026*); NVIDIA-Thinking Machines 1 GW deal (NVIDIA, Mar 2026). * = press-reported.

SO WHAT

China leads open-model downloads

Washington and NVIDIA want a US option

Reflection, TML can fill the gap

Six signposts to watch

Signpost	Watch	Bull reading	Bear reading
GPU prices	H100 index, list hikes	Hikes stick; index holds	Index below $2/hr
Live power	GW energised vs contracted	Sites on schedule	Energisation slips
Who pays	Customer mix, prepayments	Investment-grade buyers	VC-funded labs dominate
Cost of debt	GPU-loan spreads, converts	Spreads keep tightening	Refinancing at higher rates
Useful life	Depreciation policy changes	Old GPUs keep renting	Life cuts, impairments
Neolab rounds	Talks that close, step-ups	Rounds close at stated prices	Down rounds, departures

SOURCES

Where the numbers come from

Company filings and releases

CoreWeave quarterly releases Q1 2024-Q2 2026, FY2025 10-K, 8-Ks (DDTL 4.0), 17 Sep 2026 pricing release; Nebius shareholder letters Q4 2025-Q2 2026, nebius.com/prices; Crusoe, IREN, Applied Digital, Cerebras releases; Microsoft, Alphabet, Amazon, Meta earnings releases; Anthropic and OpenAI funding posts; NVIDIA releases, blogs and Nemotron model cards; AMD World Labs release and 8-K

Research and data

Epoch AI (lab revenue, inference prices, open-closed gap); Artificial Analysis index; SemiAnalysis, Silicon Data and Ornn GPU indices; LBNL Queued Up 2025 and Dec 2024 data-centre report; Uptime Institute; Meta Llama 3 paper; ByteDance MegaScale; Cottier et al. 2024; stockanalysis.com

Press-reported (labelled on slides)

Bloomberg, CNBC, The Information, Axios, FT, NYT, WSJ, TechCrunch, Forbes, Fortune, DCD, Utility Dive, POWER, TNW, SiliconANGLE, Tech.eu, PYMNTS, Capacity, TipRanks

Investor analyses

ellipsis Ventures; Radical Ventures; J. Seidel neolab analysis; Rothschild & Co Redburn (via press)

Full citations sit in each exhibit footer and in the speaker notes.